



ABSORPTION OF CARBOHYDRATES

Digestion of Carbohydrates

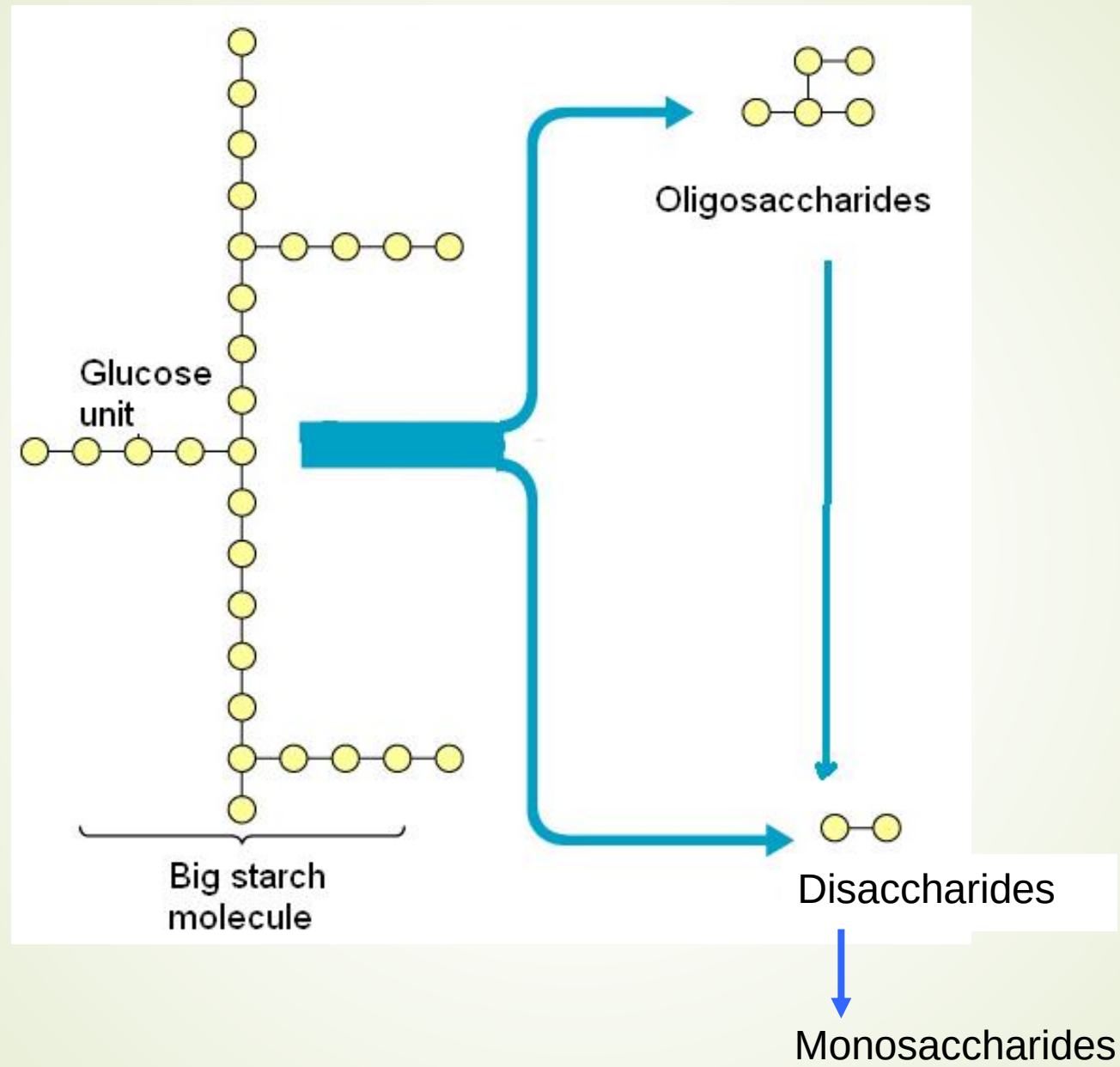
Cooking helps in breaking of glycosidic linkages in polysaccharides and thus makes the digestion process easier

In the diet carbohydrates are available as polysaccharides (starch, glycogen), and as disaccharides (sucrose and lactose). These are hydrolysed to monosaccharide units in gastro intestinal tract.

Digestion in mouth by the salivary alpha-amylase. The gastric HCl inhibits the action of salivary amylase.

In pancreatic juice another **alpha-amylase which will hydrolyse alpha-1,4 glycosidic linkages randomly, so as to produce smaller subunits like maltose, isomaltose, dextrans and branched or unbranched oligosaccharides.**

The intestinal juice (succus entericus) and brush border of intestinal cells contain **sucrase, maltase, isomaltase and lactase, The monosaccharides are then absorbed.**



Lactose Intolerance

Lactase hydrolyzes lactose to glucose and galactose.

Lactase is present in the brush border of enterocytes.

Deficiency of lactase leads to lactose intolerance. In this condition, lactose accumulates in the gut. Irritant diarrhea and flatulence are seen.

The condition should be recognized and treated immediately in newborns by giving lactose free formula diet instead of milk.

There may be congenital (primary) or acquired (secondary) causes. As age **advances**, lactase enzyme will be lost.

Acquired lactose intolerance sudden change into a milk based diet. Lactase is an **inducible enzyme**. Milk could be withdrawn temporarily, the diarrhoea will be limited.

Curd is an effective treatment, lactobacilli in curd contains the enzyme lactase. Lactase is abundantly seen in **yeast**, which is also used in treatment.

Deficiency of lactase (**alactasia**) is found in Asian population. Adults have low lactase activity than children (**hypolactasia**). So elderly people develop lactose intolerance when more milk is consumed.

Absorption of Carbohydrates

Only **monosaccharides** are absorbed by the intestine. Minute quantities of disaccharides that may be absorbed, are immediately eliminated through kidneys.

Absorption rate

Galactose > glucose > fructose

Absorption of Glucose

Glucose is polar, it cannot diffuse through the lipid bilayer of cell membrane. Hence glucose has transporters, transmembrane proteins spanning the width of the membrane.

Absorption from intestinal lumen into intestinal cell is by **co-transport** mechanism (secondary active transport)

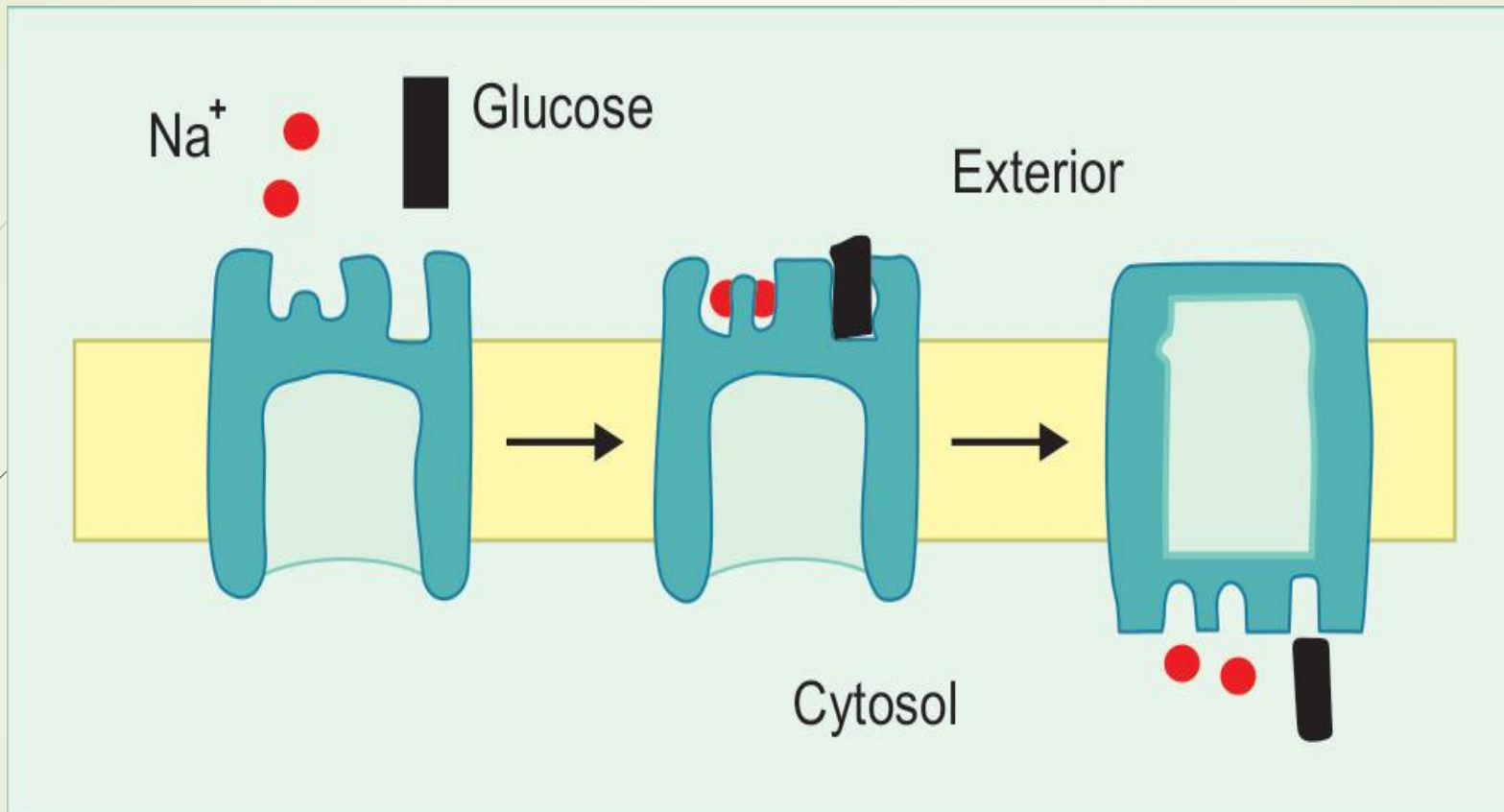
Sodium Dependent Glucose Transporter (SGluT)

Dietary Carbohydrates and their Digestion

Name of CHO	Dietary source	Digestive Enzyme	Location	Products
Starch	Cereals – Rice, Wheat	Alpha amylase	Salivary amylase Pancreatic amylase	Dextrins Maltose Isomaltose
Glycogen	Meat	Alpha amylase	Pancreatic amylase	Maltose and isomaltose
Lactose	Milk sugar	Lactase	Intestinal brush border	Glucose and galactose
Sucrose	Cane sugar	Sucrase	Intestinal brush border	Glucose and fructose

Dietary Carbohydrates and their Digestion, Continued...

Name of CHO	Dietary source	Digestive Enzyme	Location	Products
Maltose and iso-maltose	Hydro-lysis of starch	Maltase – isomaltase complex	Intestinal brush border of jejunum.	Glucose
Mono-saccharides	Fructose, pentoses Fruits and honey	No digestion		Fructose, pentoses
Fiber Cellulose and hemicellulose, pectin etc	Plant polysaccharides	No digestion	For bulk of stools and bowel movements	Fermented by intestinal bacteria. Can cause flatulence



Sodium dependent glucose transporter (SGLuT). Sodium and glucose cotransport system at luminal side; sodium is then pumped out. **Energy is used indirectly**

This type of **co-transport** is also utilised to reabsorb glucose from kidney tubules. Transporter in **intestine** is **SGluT-1** and transporter in the **kidney** is called **SGluT-2**.

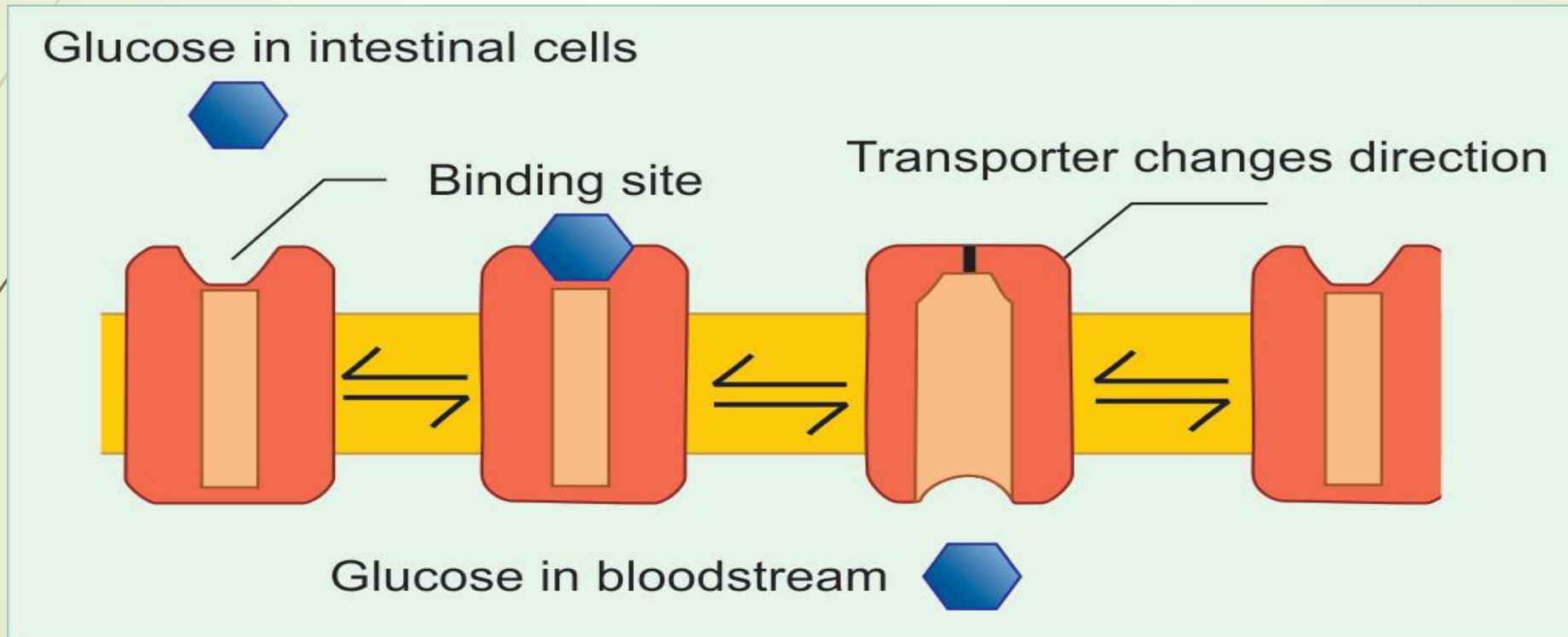
Common treatment for diarrhea is **oral rehydration fluid**. It contains glucose and sodium. Presence of glucose in oral rehydration fluid allows uptake of sodium to replenish body sodium chloride.

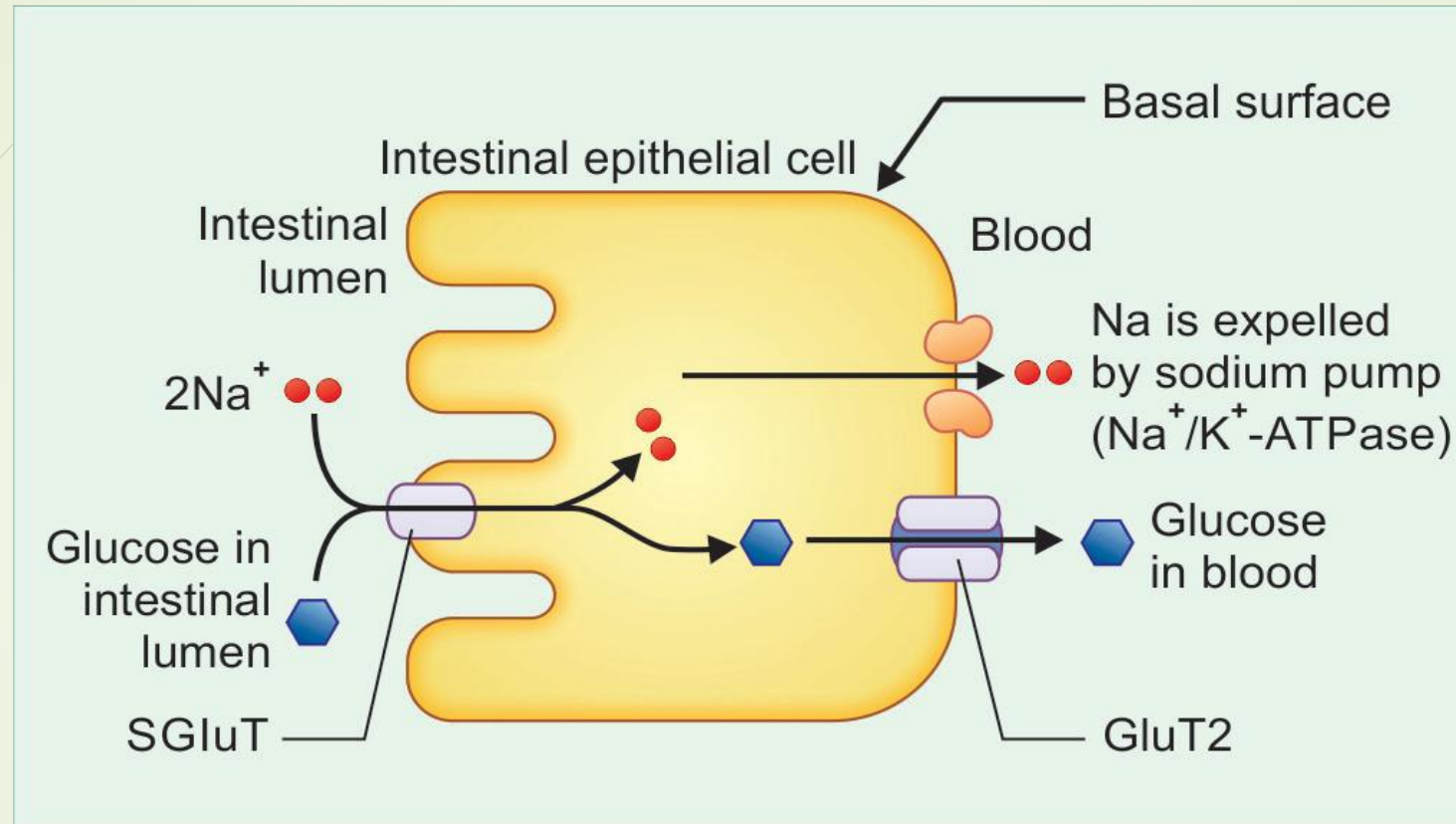
Another Uniport System Releases Glucose into Blood

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Glucose Transporter Type 2 (GluT2)

This transporter is not dependent on sodium, but it is a **uniport, facilitated diffusion** system

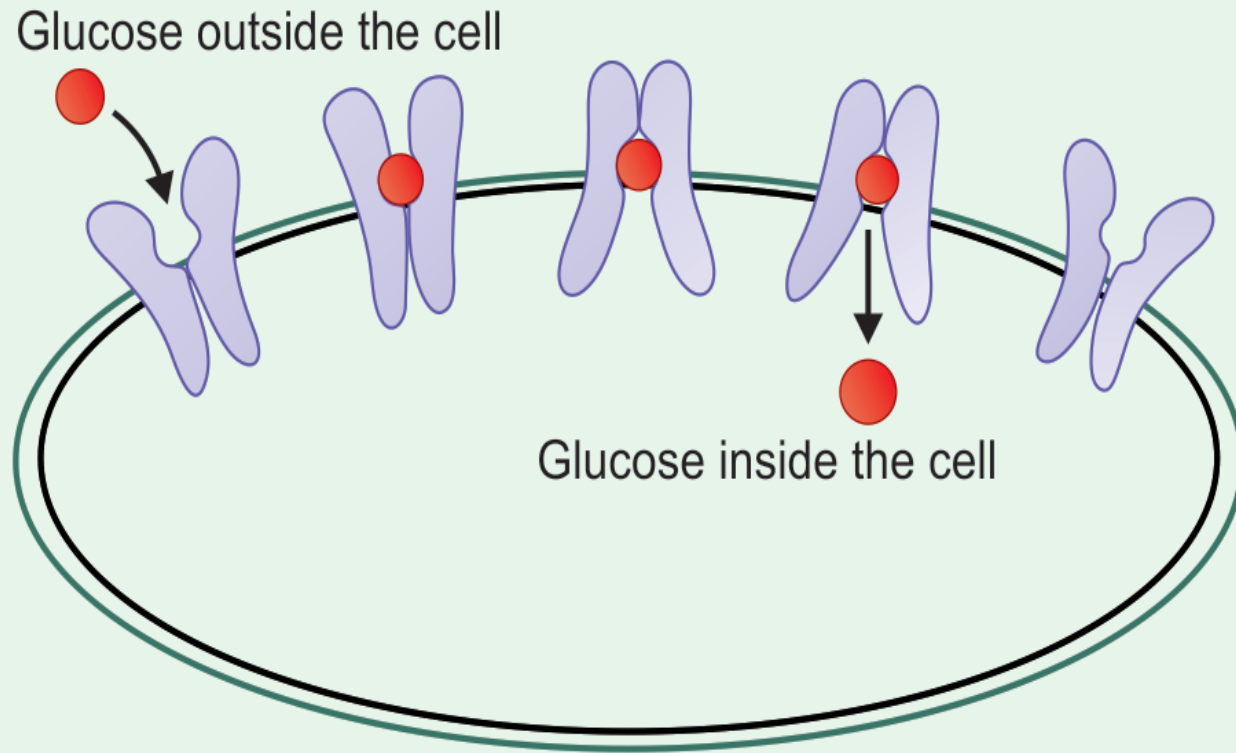




Intestinal absorption of glucose. At the intestinal lumen, absorption is by sodium dependent glucose transporter (SGLuT) and at the blood vessel side, absorption is by glucose transporter 2 (GluT2).

GluT2 transports glucose into cells when blood sugar level is high. So in the well-fed state, glucose is taken up by liver and deposited as glycogen.

This mechanism also enables the pancreas to monitor the glucose level and adjust the rate of insulin secretion.



Glucose transporter 4 (GluT4)
Glucose transport in cells

- ❖ The GluT4 is the major glucose transporter in **skeletal muscle and adipose tissue**.
- ❖ GluT4 is under the control of **insulin**. But other glucose transporters are not under the control of insulin.

Glucose Transporter 4 in Skeletal Muscle Heart Muscle and Adipose Tissue

During exercise, muscle accounts for 80% of body's glucose utilisation.

But at basal metabolic state, brain utilises 60% of glucose oxidised in the body.

The GluT4 is under control of insulin.

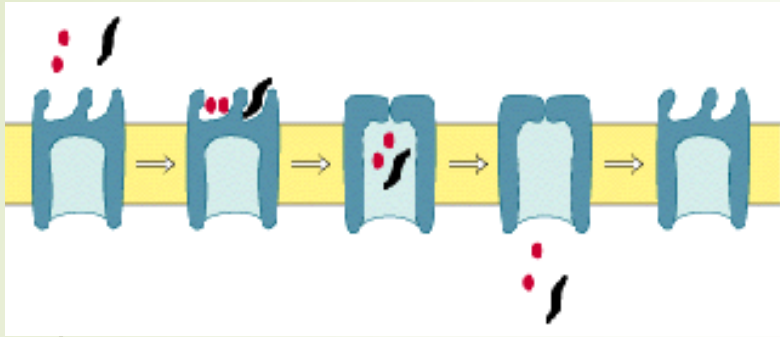
Other glucose transporters are not under the control of insulin.

Insulin induces intracellular GluT4 molecules to move to the cell membrane and thus increases glucose uptake.

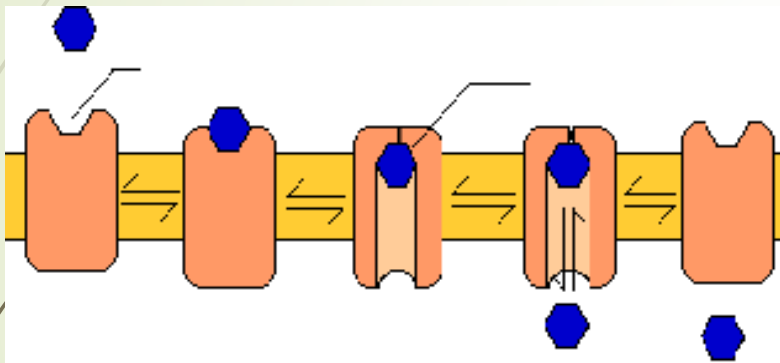
In Type 2 diabetes mellitus, insulin resistance is seen in muscle and fat cells.

In diabetes, entry of glucose into muscle is only half of normal cells.

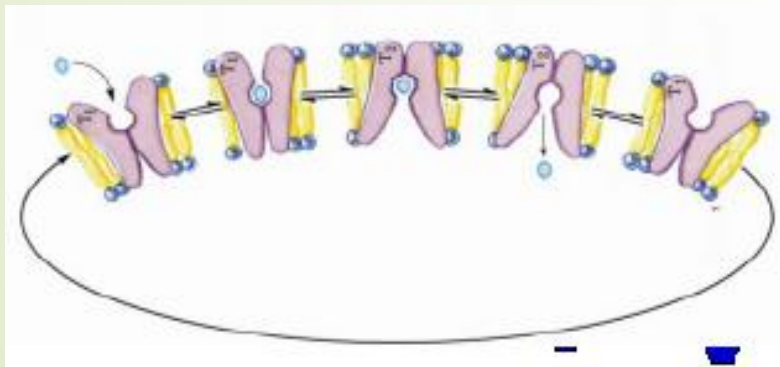
Reduced recycling of membrane bound GluT4



SGLuT. Sodium and glucose Co-transport at luminal side; sodium is then pumped out.



Glu2 at vascular site.

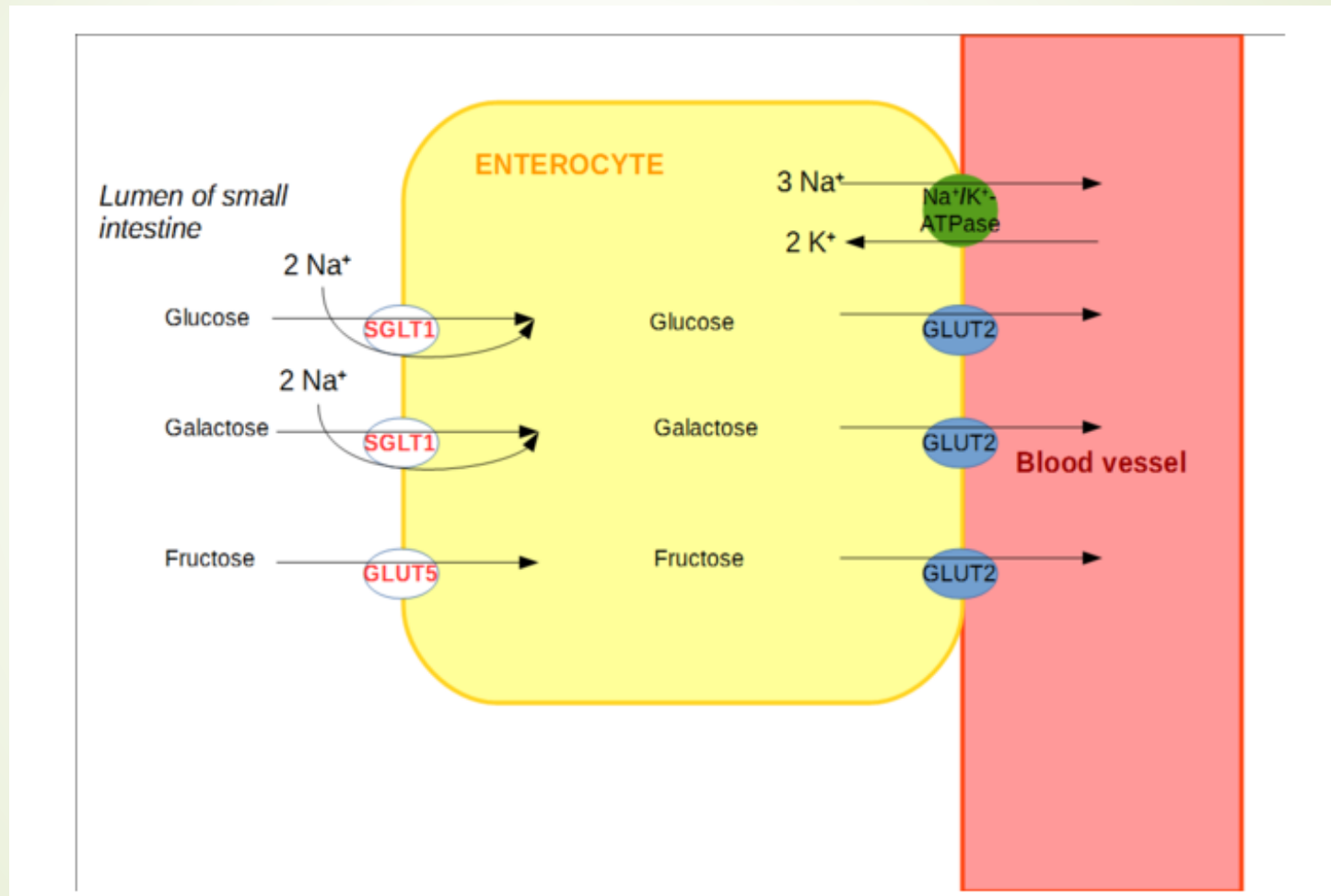


GluT4. Glucose transport in cells

Glucose Transporters

	Present in	Properties
Glut1	RBC, brain, kidney,	Glucose uptake in colon, retina, most of cells placenta
Glut2	Serosal surface of intestinal cells, liver, beta cells of pancreas	Glucose uptake in liver; glucose sensor in beta cells
Glut3	Neurons, brain	Glucose into brain cells
Glut4	Skeletal, heart	Insulin mediated muscle, adipose tissue glucose uptake
Glut5	Small intestine, testis, sperms, kidney	Fructose transporter; poor ability to transport glucose
SGLuT	Intestine, kidney	Cotransport; from lumen into cell

Absorption of Monosaccharides



GLYCOLYSIS AND GLUCONEOGENESIS

Glycolysis

(Embden-Meyerhof Pathway)

Definition: Glucose is split into two 3-carbon **pyruvate** molecules under aerobic conditions; or **lactate** under anaerobic conditions, along with production of a small quantity of energy.

Site of reactions: All the reaction steps take place in the cytoplasm.

Significance of Glycolysis Pathway

1. It is the only pathway that is taking place in all cells of the body.
2. Glycolysis is the only source of energy in erythrocytes.
3. In strenuous exercise, when muscle tissue lacks enough oxygen, **anaerobic glycolysis forms the major source of energy for muscles.**
4. The glycolytic pathway may be considered as the preliminary step before complete oxidation.
5. The glycolytic pathway provides carbon skeletons for synthesis of non-essential amino acids as well as glycerol part of fat.
6. Most of the reactions of the glycolytic pathway are reversible, which are also used for gluconeogenesis.

Clinical Importance of Glucose

A minimum amount of glucose is always required for normal functioning.

Normal fasting plasma glucose level is 70–100 mg/dL.

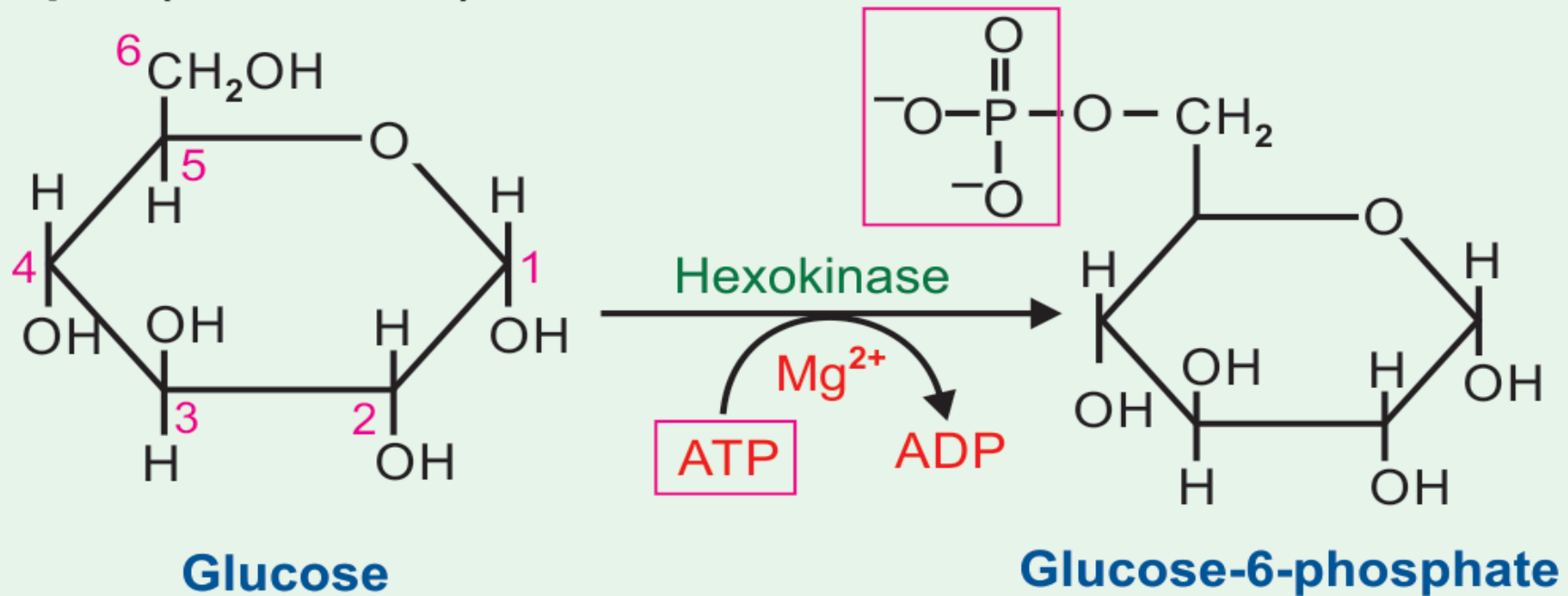
After a heavy carbohydrate meal, it rises; but in a normal person, this level is below 150 mg/dL.

Step 1 of Glycolysis

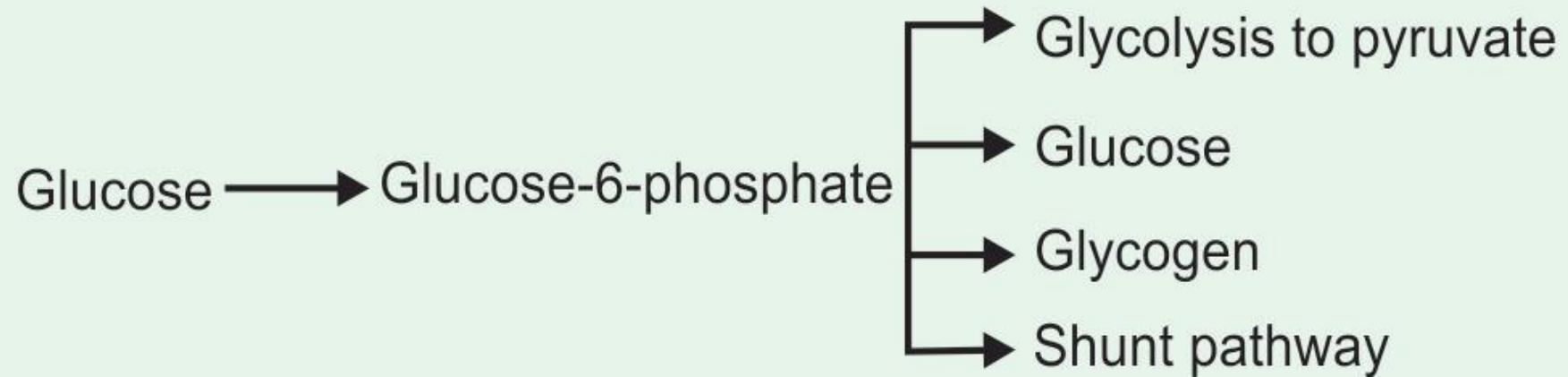
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Activation of glucose by phosphorylation.

Step 1: (Irreversible)



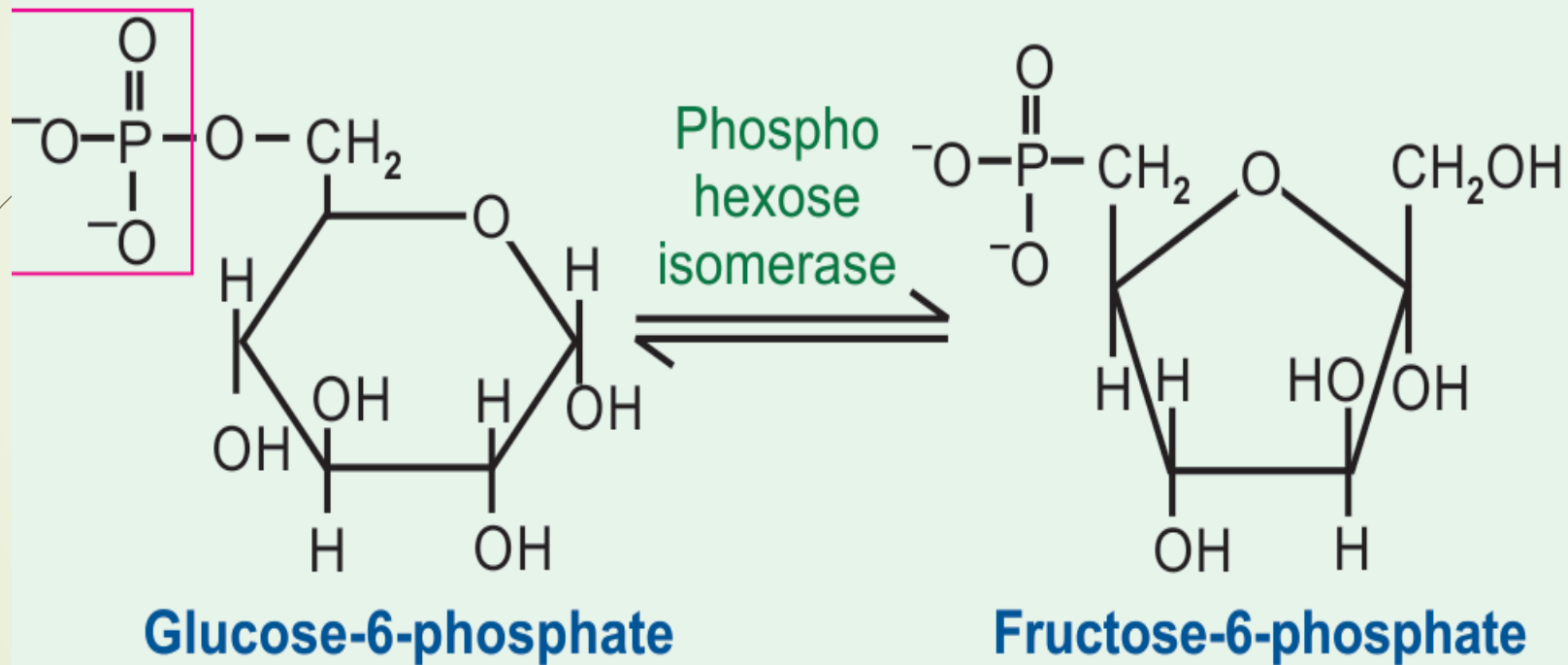
Energy Utilising Step Irreversible

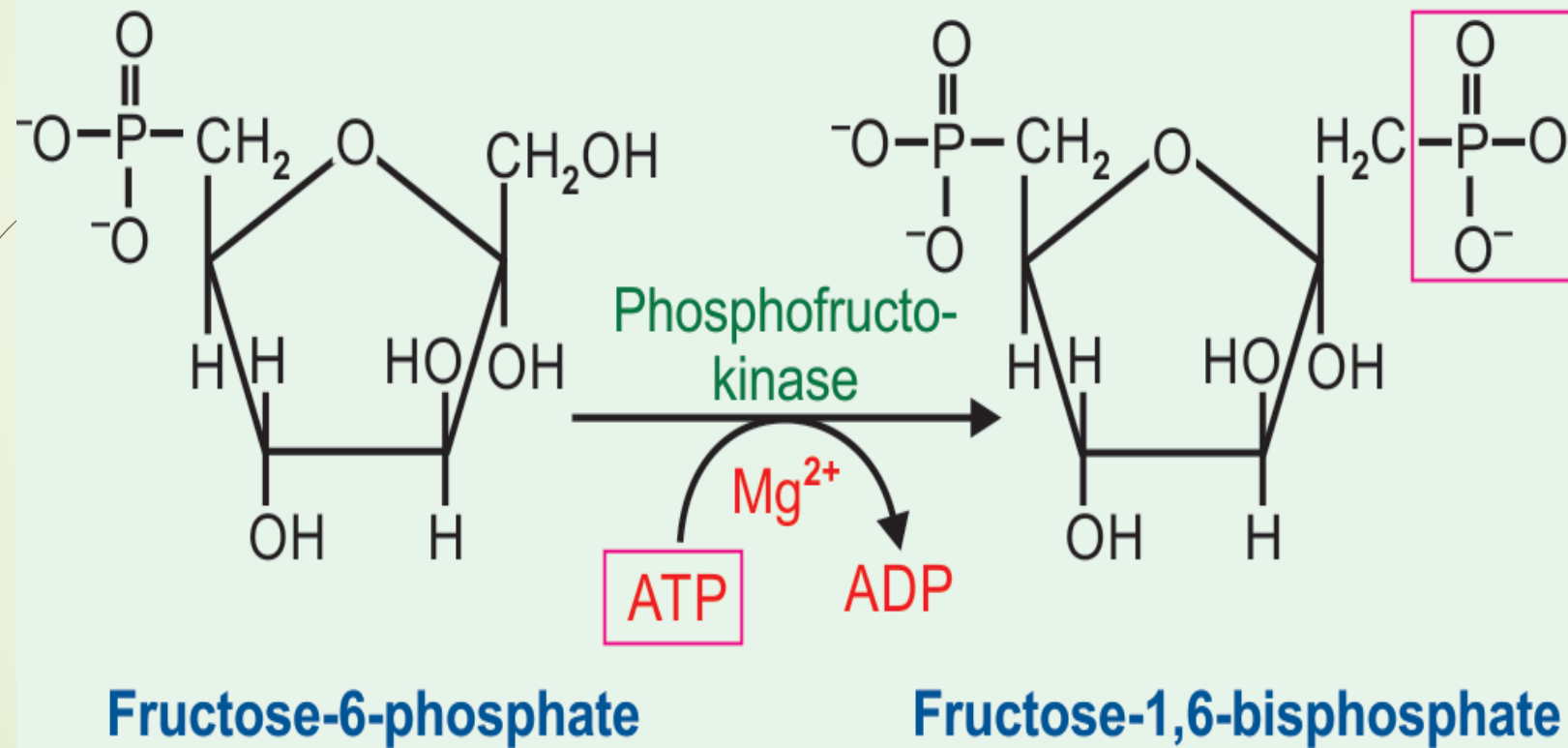


COMPARISON OF HEXOKINASE AND GLUCOKINASE

	Hexokinase	Glucokinase
Occurrence	In all tissues	Only in liver
Km value	10^{-2} mmol/L	20 mmol/L
Affinity to substrate	High	Low
Specificity	Acts on glucose, fructose and mannose	Acts only on glucose
Induction	Not induced	Induced by insulin and glucose
Function	Even when blood glucose level is low, glucose is utilised by body cells	Acts only when blood glucose level is more than 100 mg/dl; then glucose is taken up by liver cells for glycogen synthesis

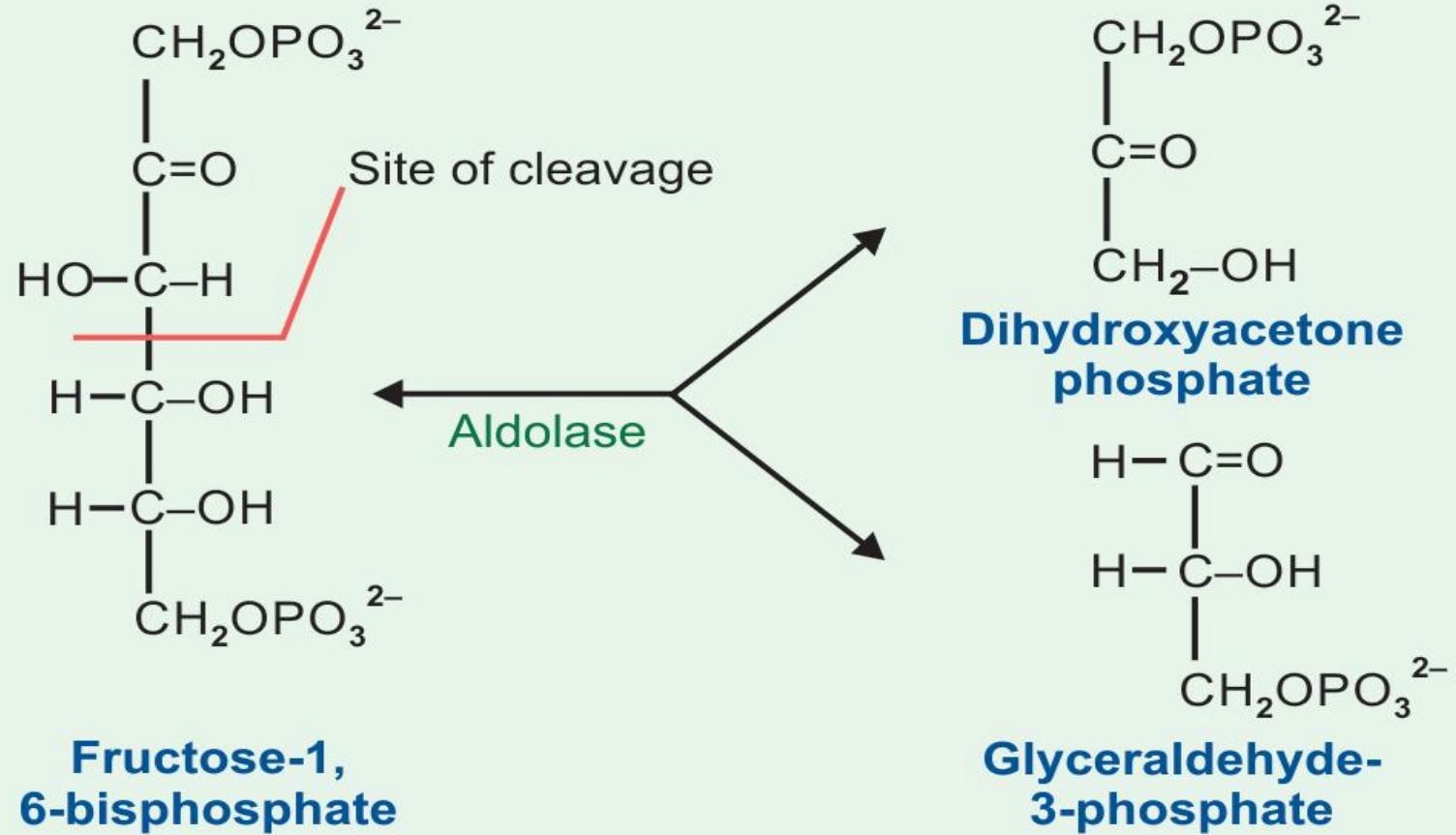
Step 2: Phosphohexose isomerase Reaction

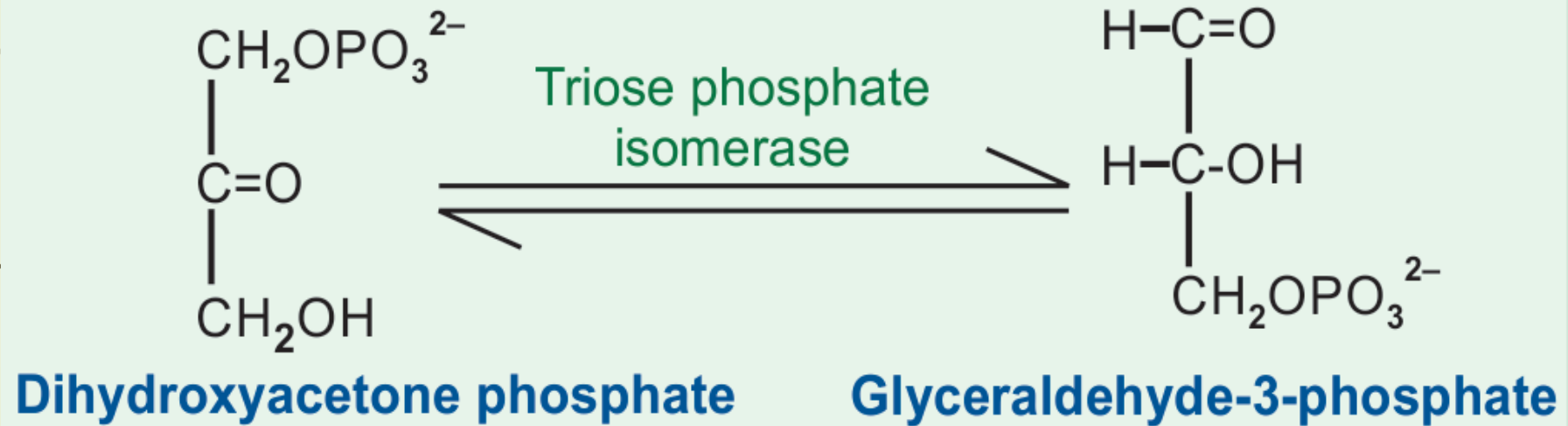


Step 3: (Irreversible)

Step 3 of Glycolysis

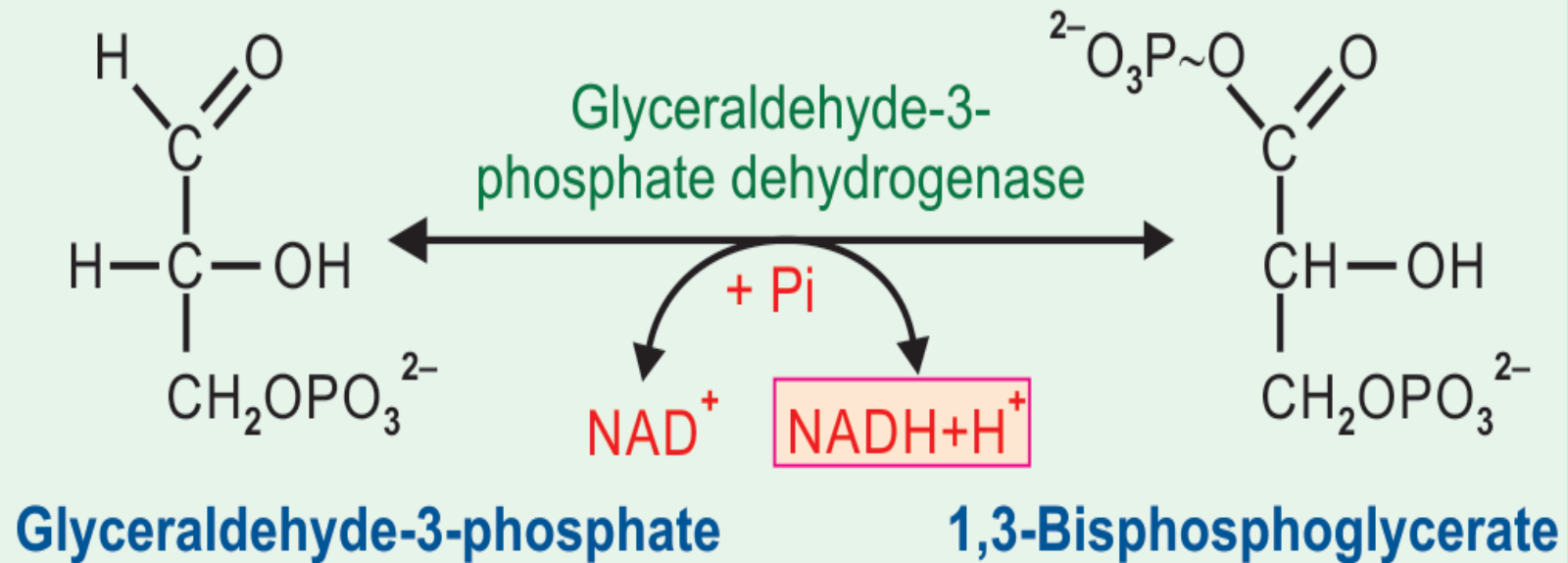
- i) Fructose-6-phosphate is further phosphorylated to **fructose1,6-bisphosphate**. The enzyme is **phosphofructo kinase**.
- ii) PFK is an allosteric, inducible, regulatory enzyme. It is an important **key enzyme** of this pathway.
- iii) This is again an activation process, the energy being derived by hydrolysis of yet another molecule of **ATP**.
- iv) This **irreversible** step is the rate limiting reaction in glycolysis.
- v) The steps 1,2 and 3 together are called as the **preparatory phase**.

Step 4: (Reversible)

Step 4A: Isomerization (Reversible)

Steps 4 and 4-A are together called the **Splitting Phase**.

Step 5: NADH Generating step (Reversible)



Reversible step.

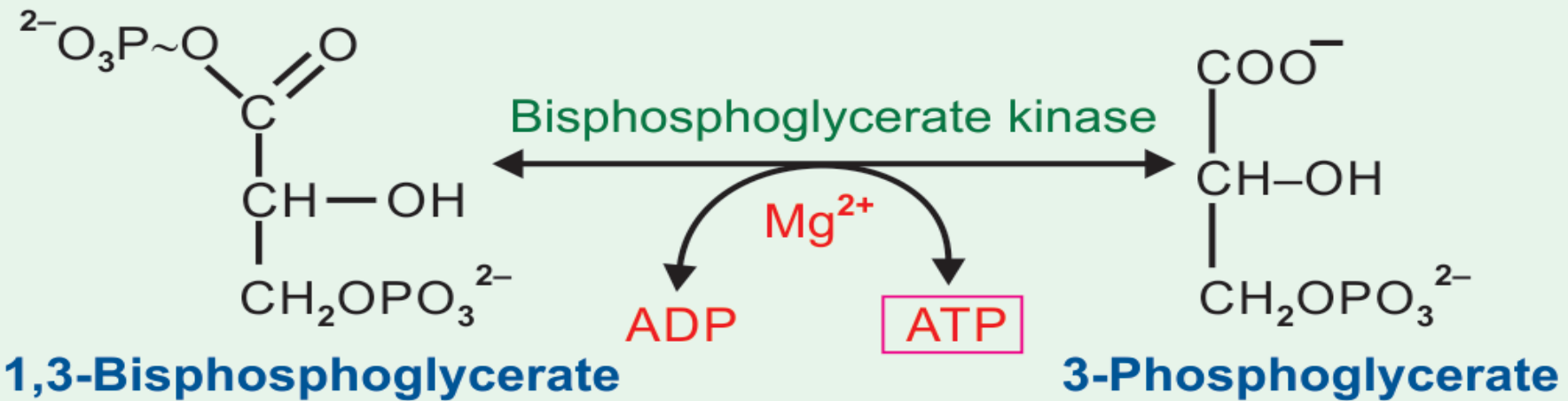
NADH generating step.

Oxidative phosphorylation

Diphosphate and Bisphosphate are Different

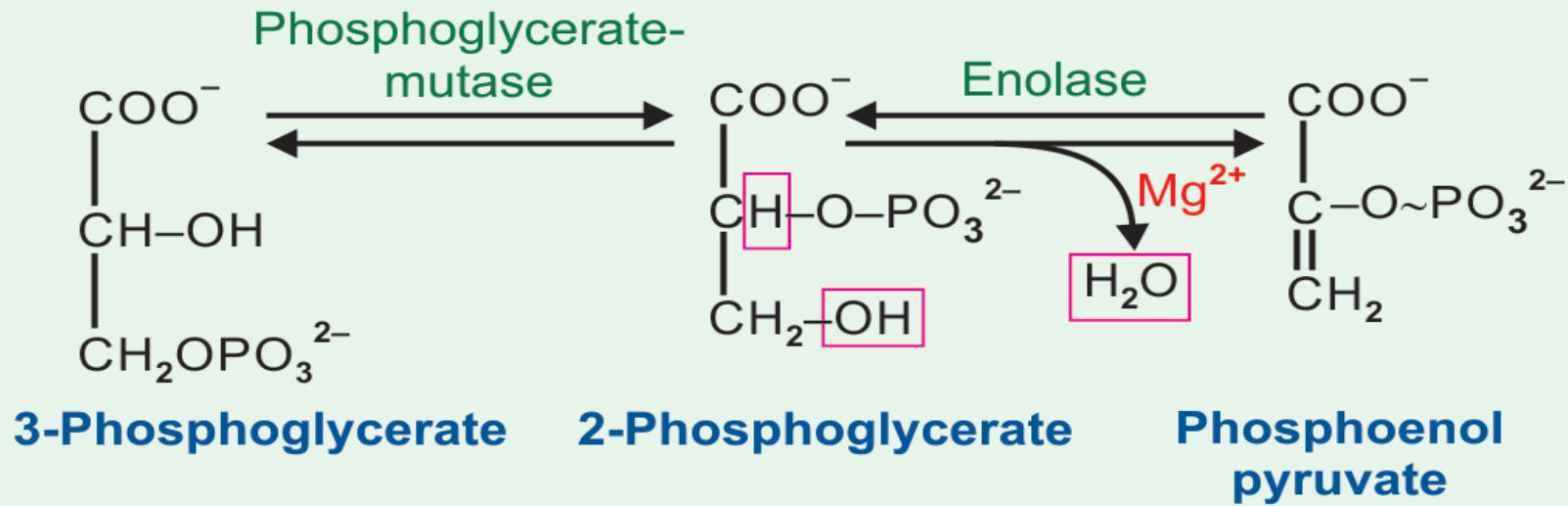
When two phosphate groups are linked together and then attached to a parent compound, it is called diphosphate, e.g. adenosine-diphosphate.

But when phosphoric acid groups are present at two different sites of the compound, it is named as bisphosphate, e.g. fructose-1,6-bisphosphate.

Step 6: ATP Generation (Reversible)

ATP generation step, Example of substrate level phosphorylation

Step 7 and 8

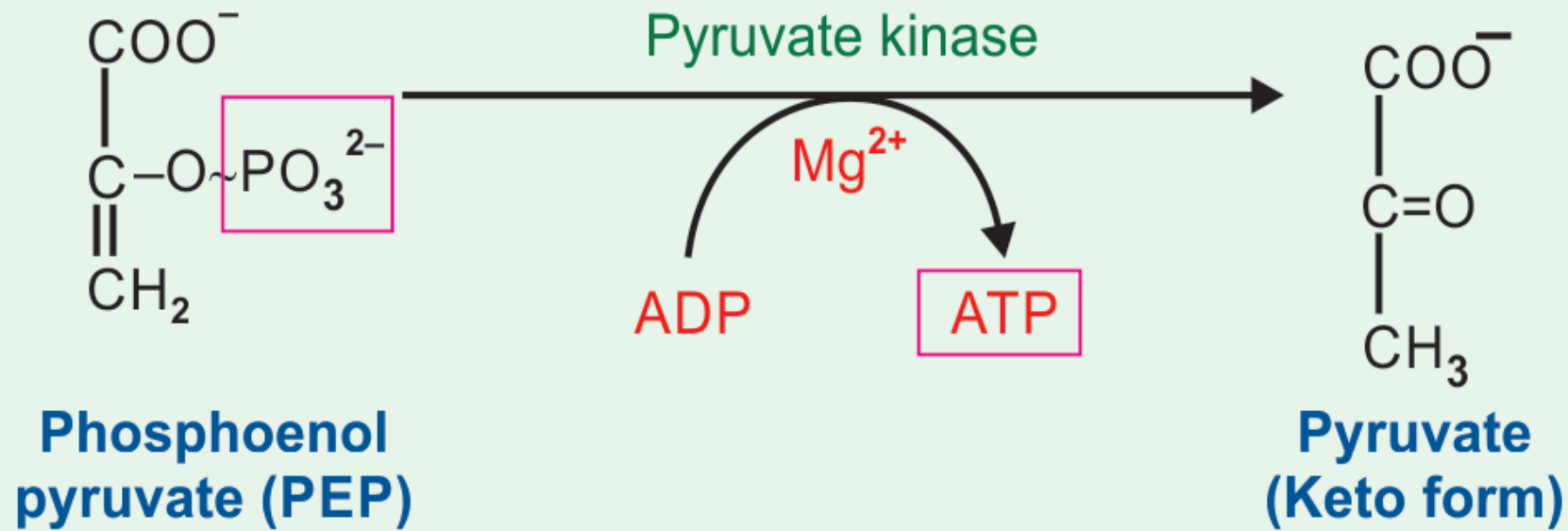


Enolase requires Mg^{++} . **Fluoride** will remove magnesium ions and inhibit this enzyme.

So when **taking blood for sugar estimation**, fluoride is added to blood.

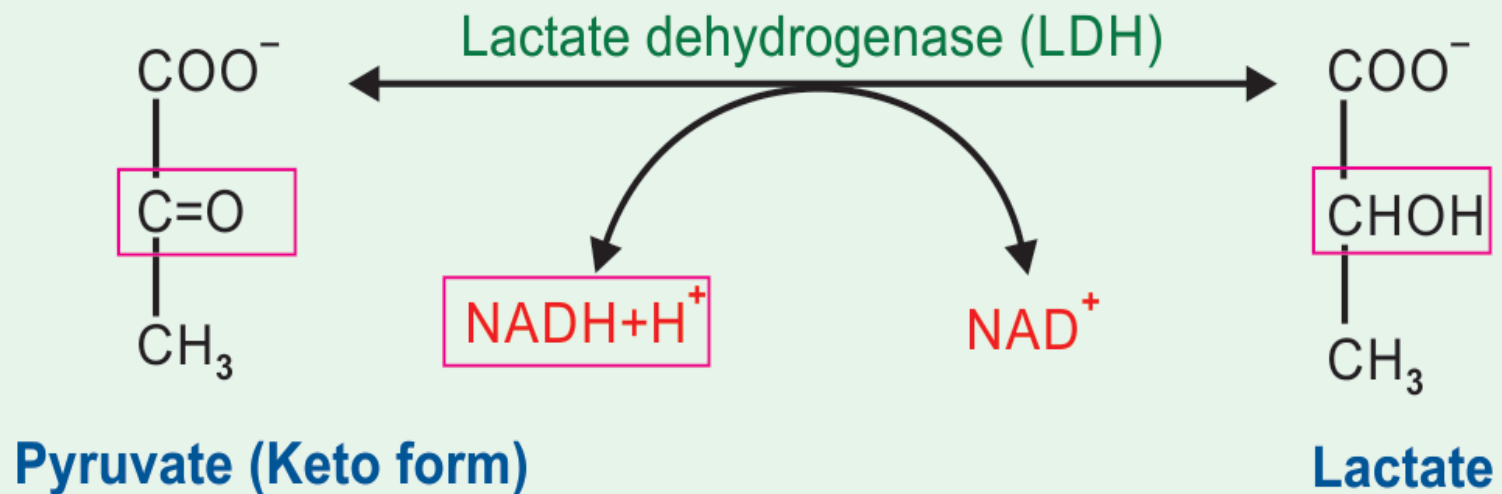
If not, glucose is metabolised by the blood cells, so that lower blood glucose values are obtained.

Step 9: ATP Production (Irreversible)

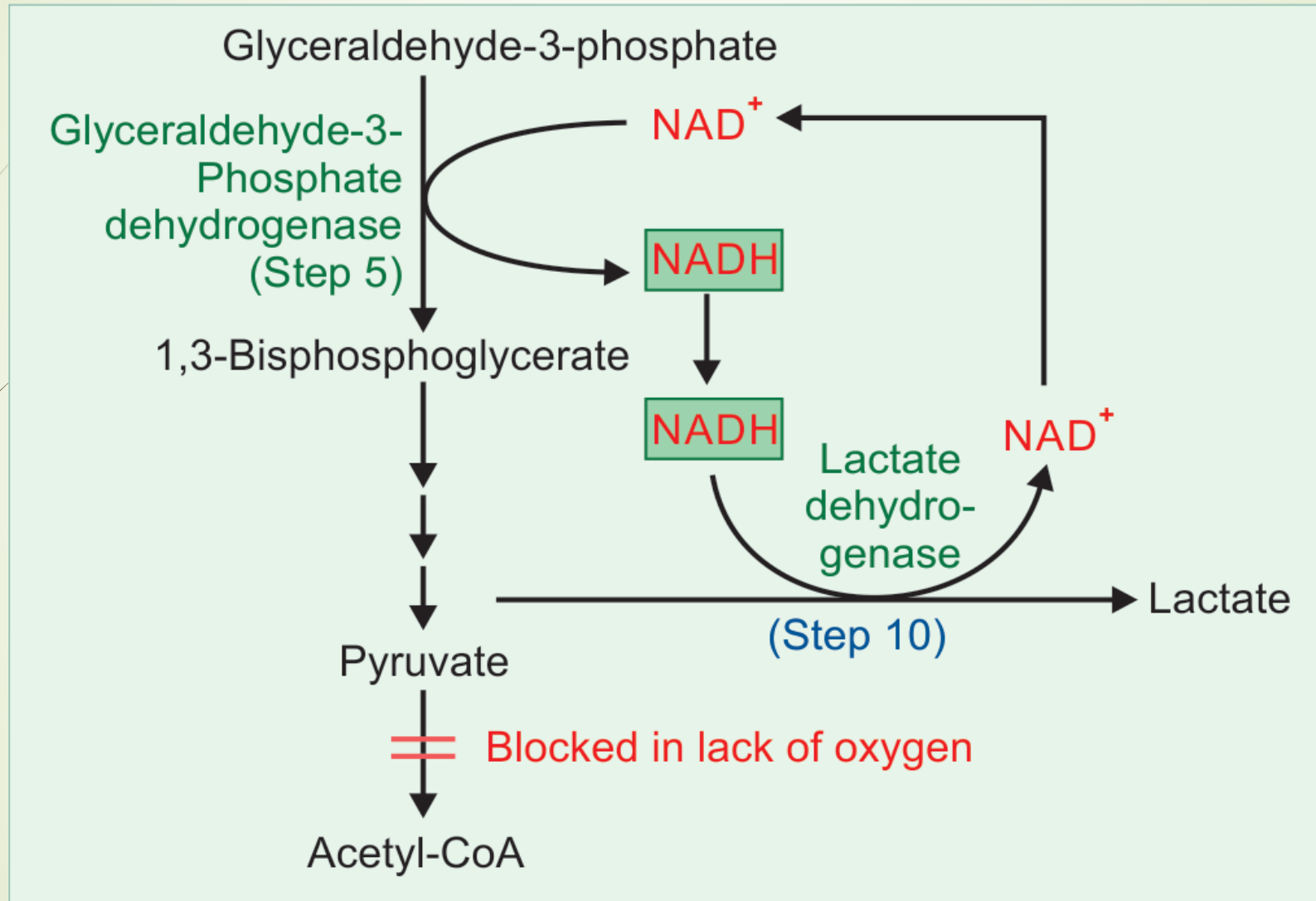


Substrate level phosphorylation

Step 10: Lactate Dehydrogenase Reaction (Reversible)



LDH has 4 subunits and 5 iso-enzymes. **The cardiac iso-enzyme** of LDH (H4) will Increase in **myocardial infarction**



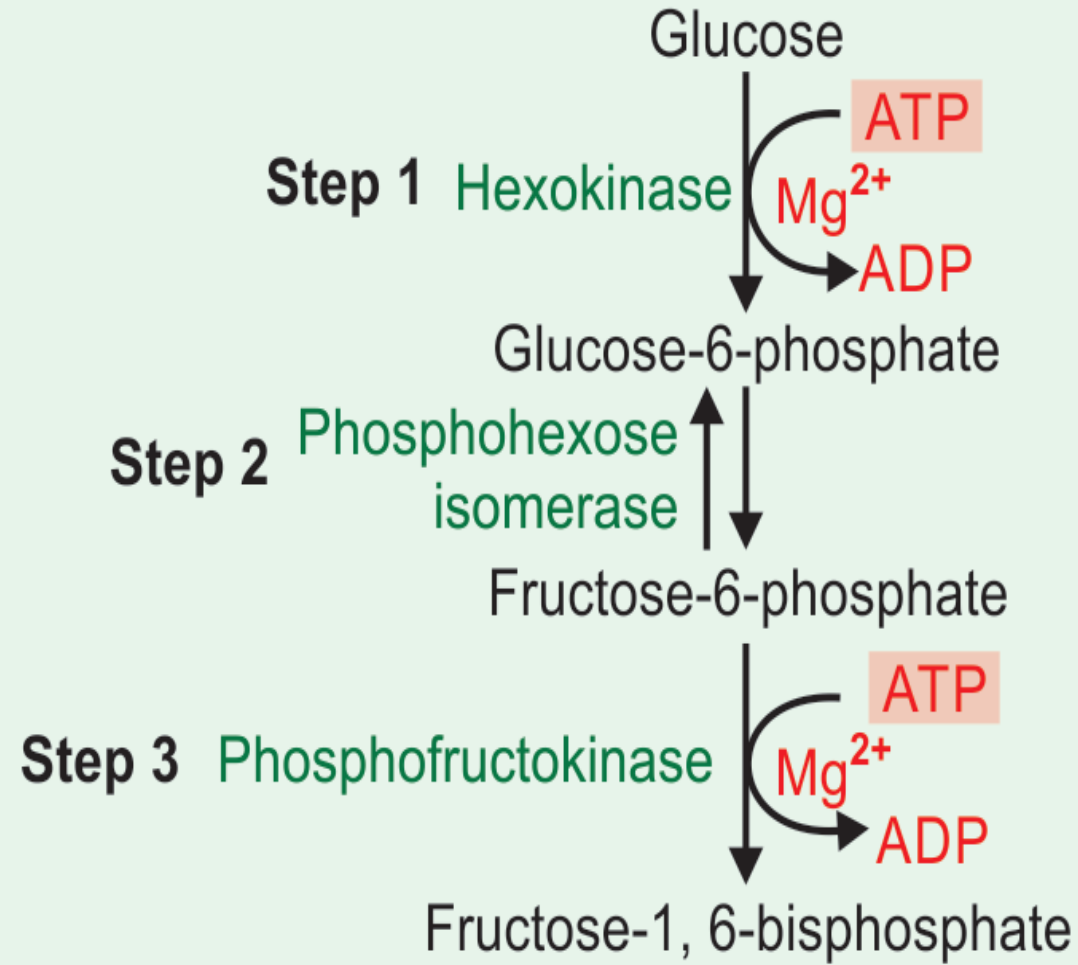
Steps 5 and 10 are Coupled

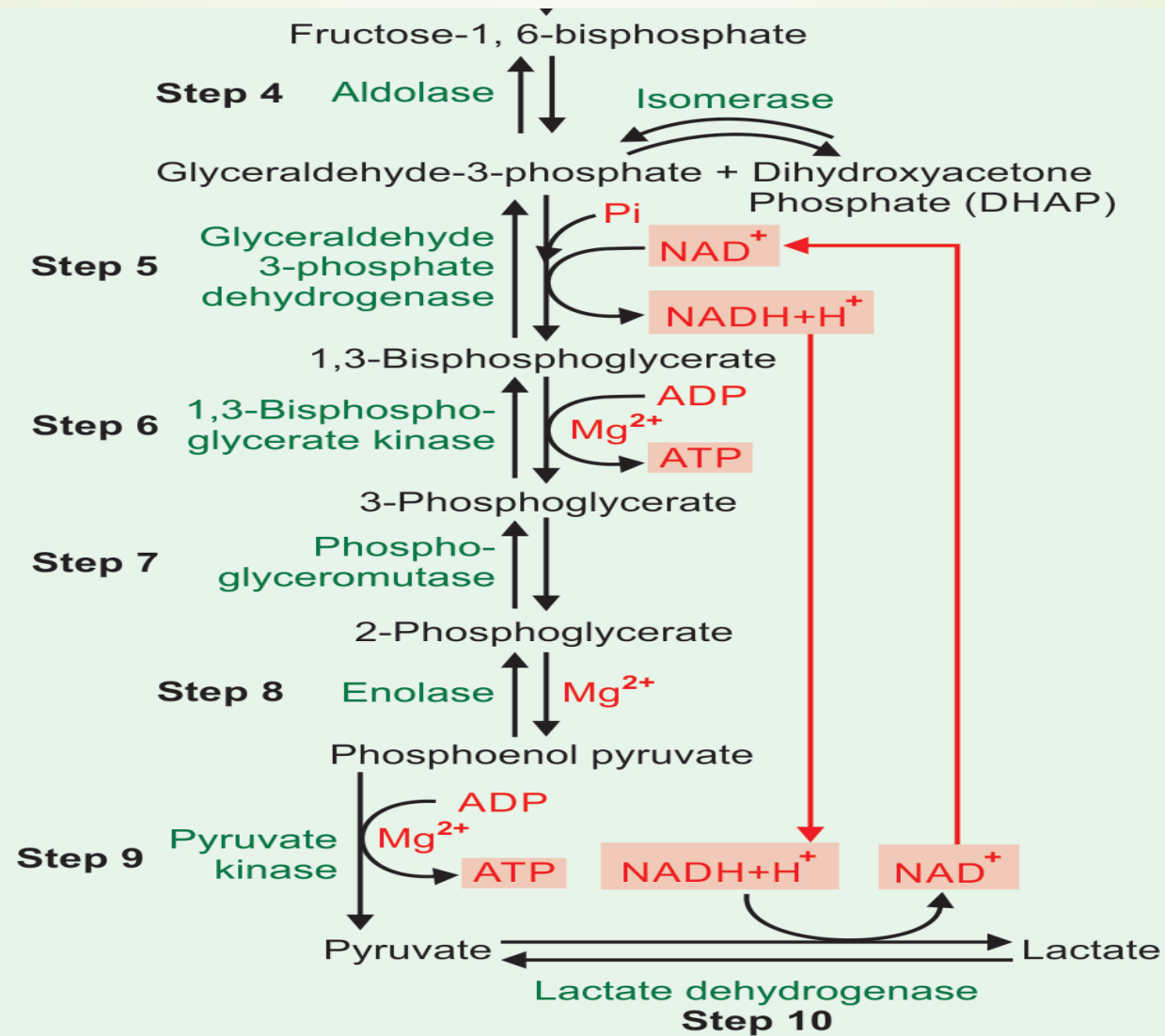
Significance of lactate production

In the actively contracting muscles, there is comparative lack of oxygen.

In anaerobiosis the major pathway of Utilisation of pyruvate is thus blocked.

Lactate formation is necessary for re-conversion of NADH to NAD⁺ in anaerobic conditions





Insulin favours glycolysis

Insulin activates key glycolytic enzymes (glucokinase, phosphofructokinase and pyruvate kinase)

Glucocorticoids inhibit glycolysis and favours gluconeogenesis

Phospho fructo kinase (PFK) (step 3) rate-limiting enzyme for glycolysis pathway

PFK is an allosterically regulated enzyme

ATP is allosteric inhibitor

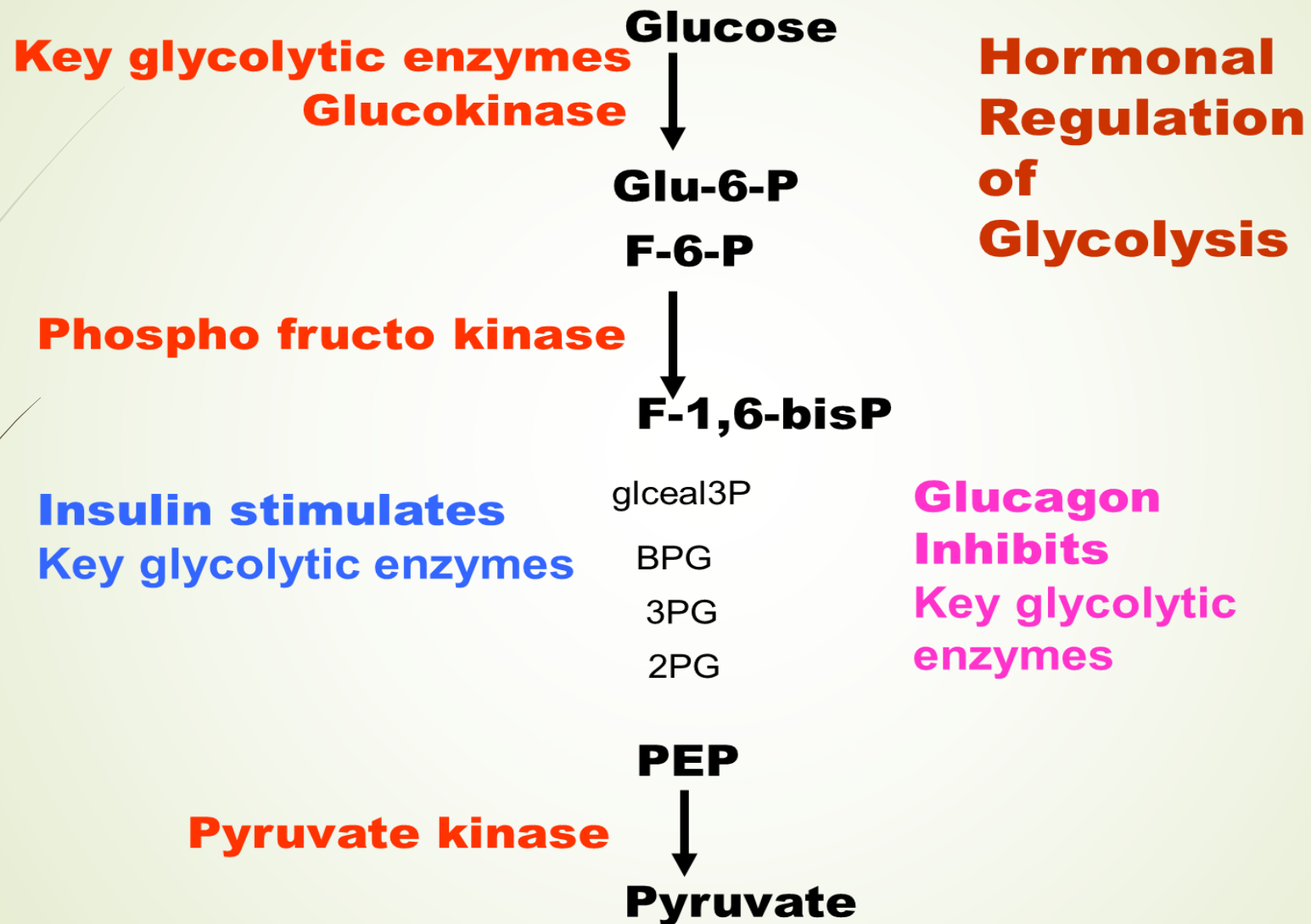
Citrate is another inhibitor

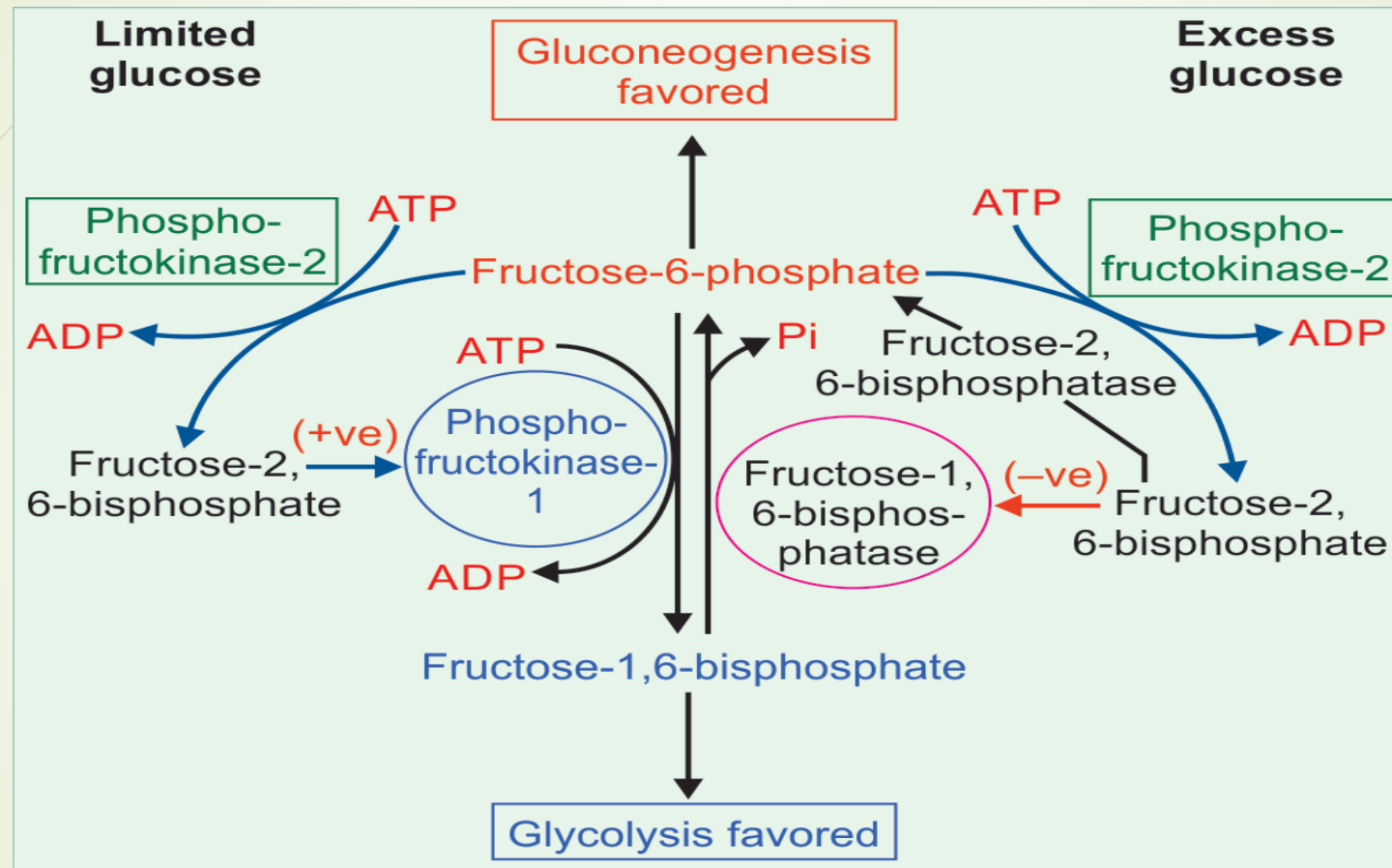
AMP acts as an allosteric activator

Fructose-6-phosphate increases activity

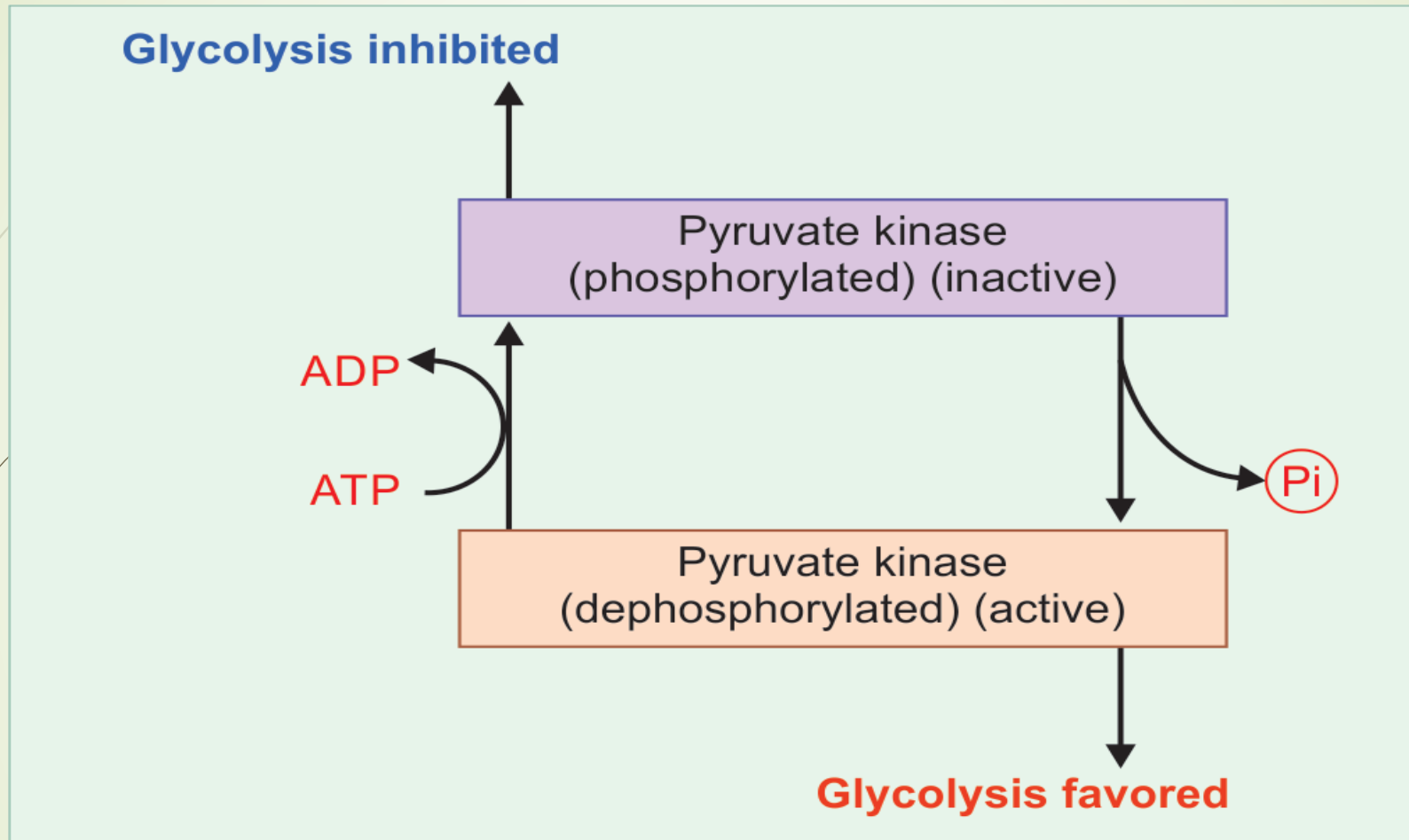
Regulatory enzymes of glycolysis

Enzyme	Activation	Inhibition
HK		G-6-P
GK	Insulin	Glucagon
PFK	Insulin, AMP F-6-P, PFK-2 F-2,6-BP	Glucagon, ATP Citrate, Low pH Cyclic AMP
PK	Insulin, F-1,6-BP	Glucagon, ATP Cyclic AMP
PDH	CoA , NAD	Acetyl CoA, NADH



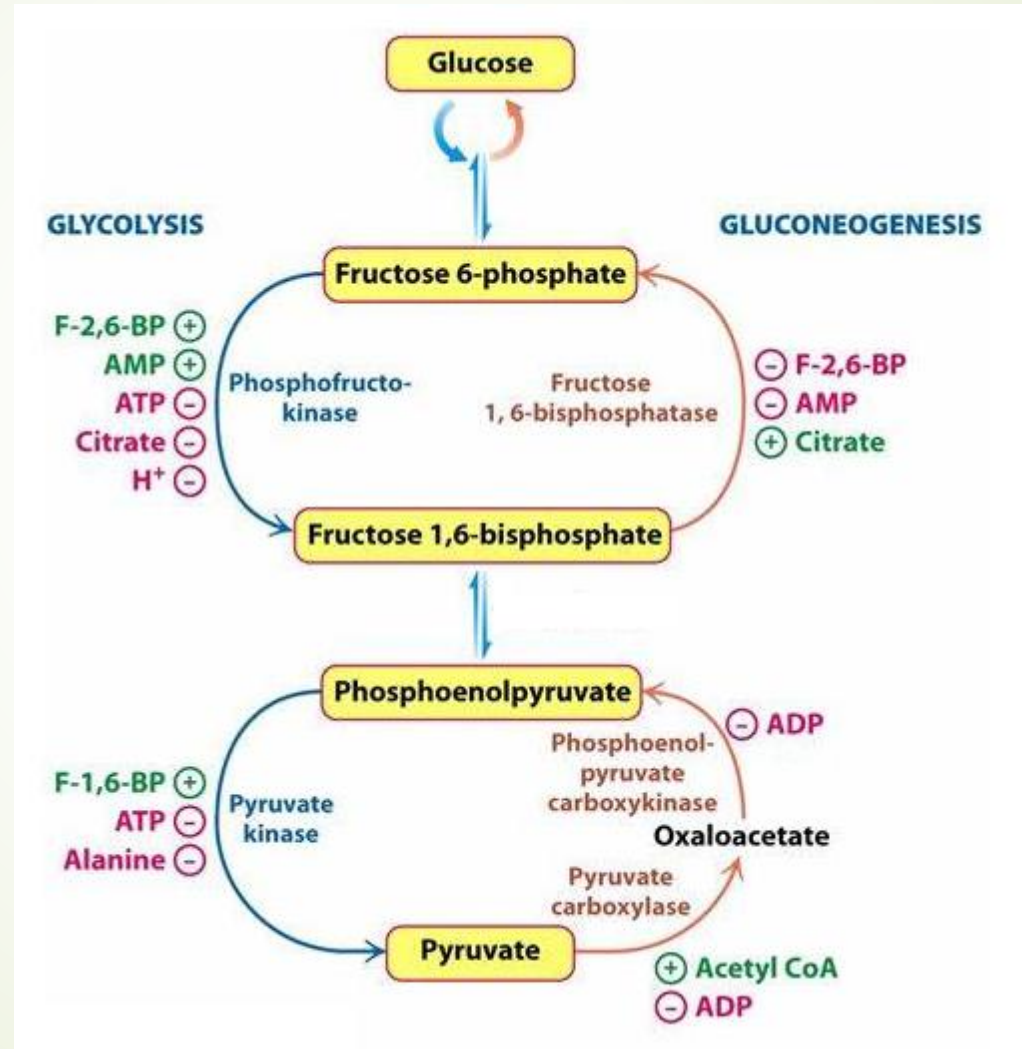


Reciprocal regulation of PFK-2 and fructose-2,6-bisphosphatase by phosphorylation.



Covalent modification of pyruvate kinase reaction; this is similar to phosphofructokinase-2 (PFK-2).

Summary of Regulation of Glycolysis



Energy yield (number of ATP generated) per molecule of glucose in the glycolytic pathway, under anaerobic conditions (Oxygen deficiency)

Step	Enzyme	Source	gained per glucose mol
1	Hexokinase	-	Minus 1
3	Phosphofructokinase	-	Minus 1
6	1,3-bisphosphoglycerate kinase	ATP	1 x 2 = 2
9	Pyruvate kinase	ATP	1 x 2 = 2
Total = 4 minus 2 = 2			

Energy yield (number of ATP generated) per molecule of glucose in the glycolytic pathway, under aerobic conditions (oxygen is available)

Step	Enzyme	Source	No of ATP gained per glucose mol
1	Hexokinase	-	Minus 1
3	Phosphofructokinase	-	Minus 1
5	Glyceraldehyde-3-phosphate dehydrogenase	NADH	$2.5 \times 2 = 5$
6	1,3-bisphosphoglycerate kinase	ATP	$1 \times 2 = 2$
9	Pyruvate kinase	ATP	$1 \times 2 = 2$
Total = 9 minus 2 = 7			

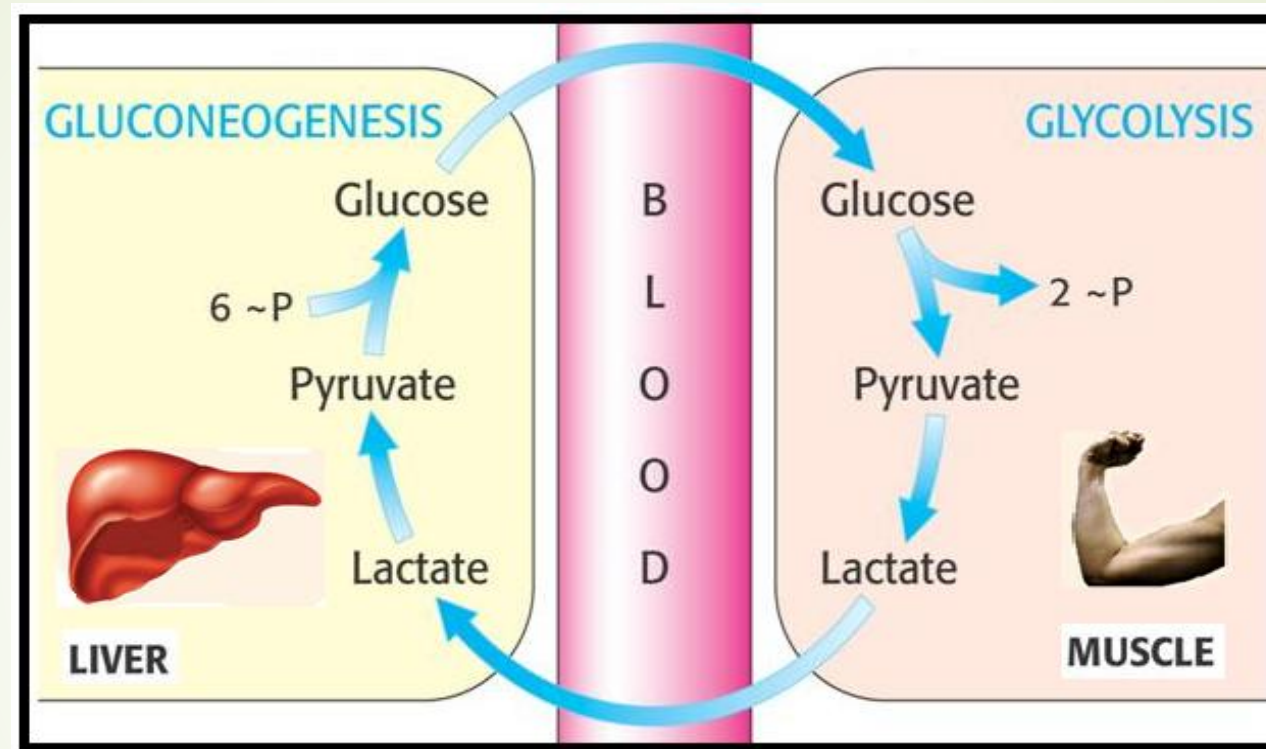
Energy yield (number of ATP generated) per molecule of glucose when it is **completely oxidised through glycolysis plus citric acid cycle, under *aerobic conditions*.**

Net generation in glycolytic pathway 9 minus 2 = 7

Generation in pyruvate dehydrogenase (pyruvate to acetyl CoA) = 5

Generation in citric acid cycle = 20

Net generation of ATP from one glucose mol = 32

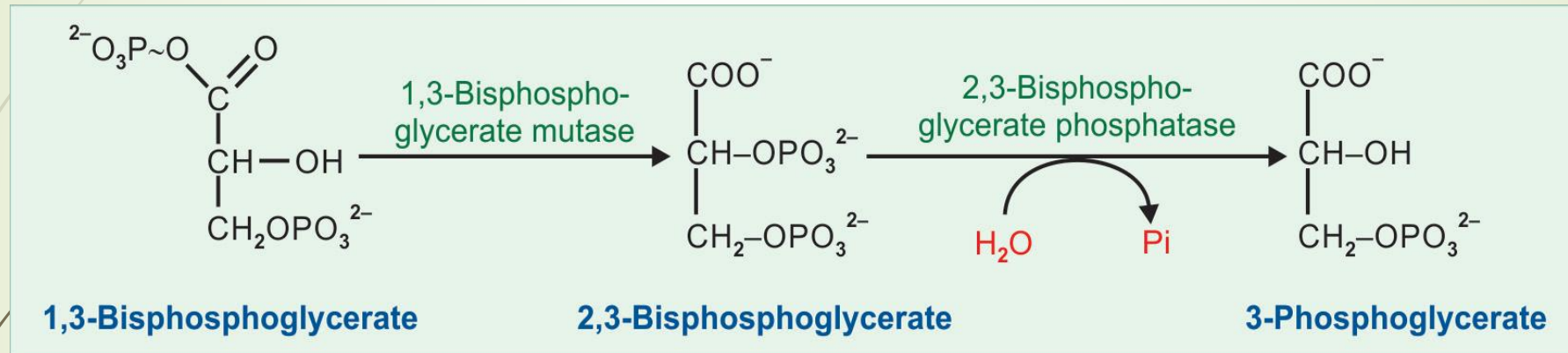


Cori's cycle.

Contracting muscle has lack of oxygen. So pyruvate is reduced to lactate. This can be reconverted to glucose in liver by gluconeogenesis.

Efficient Utilisation of Lactate.

Rapaport Leubering Cycle (BPG Shunt) in RBCs



Erythrocytes

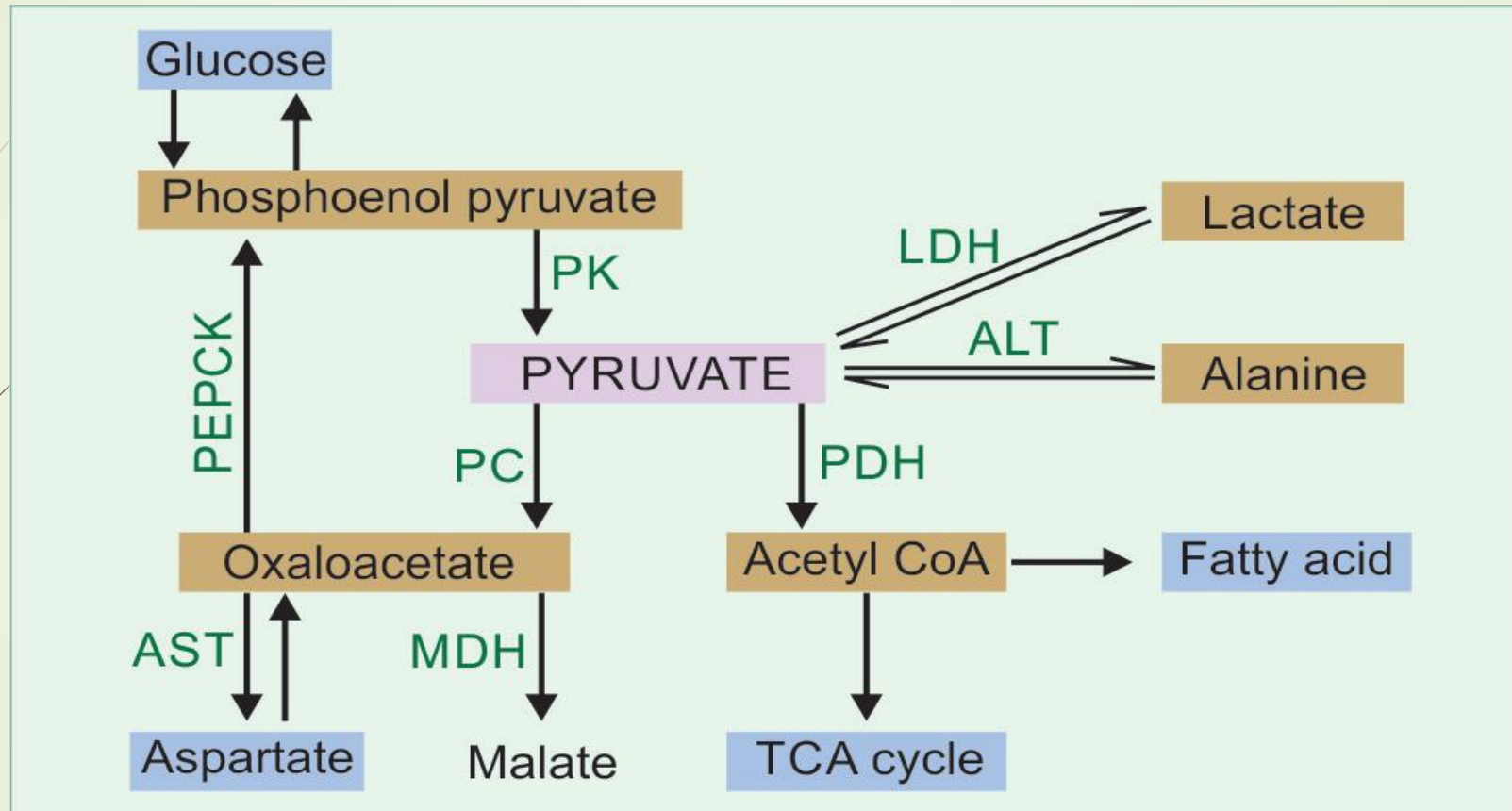
In this pathway, no ATP is generated.

Significance of Bisphosphoglycerate

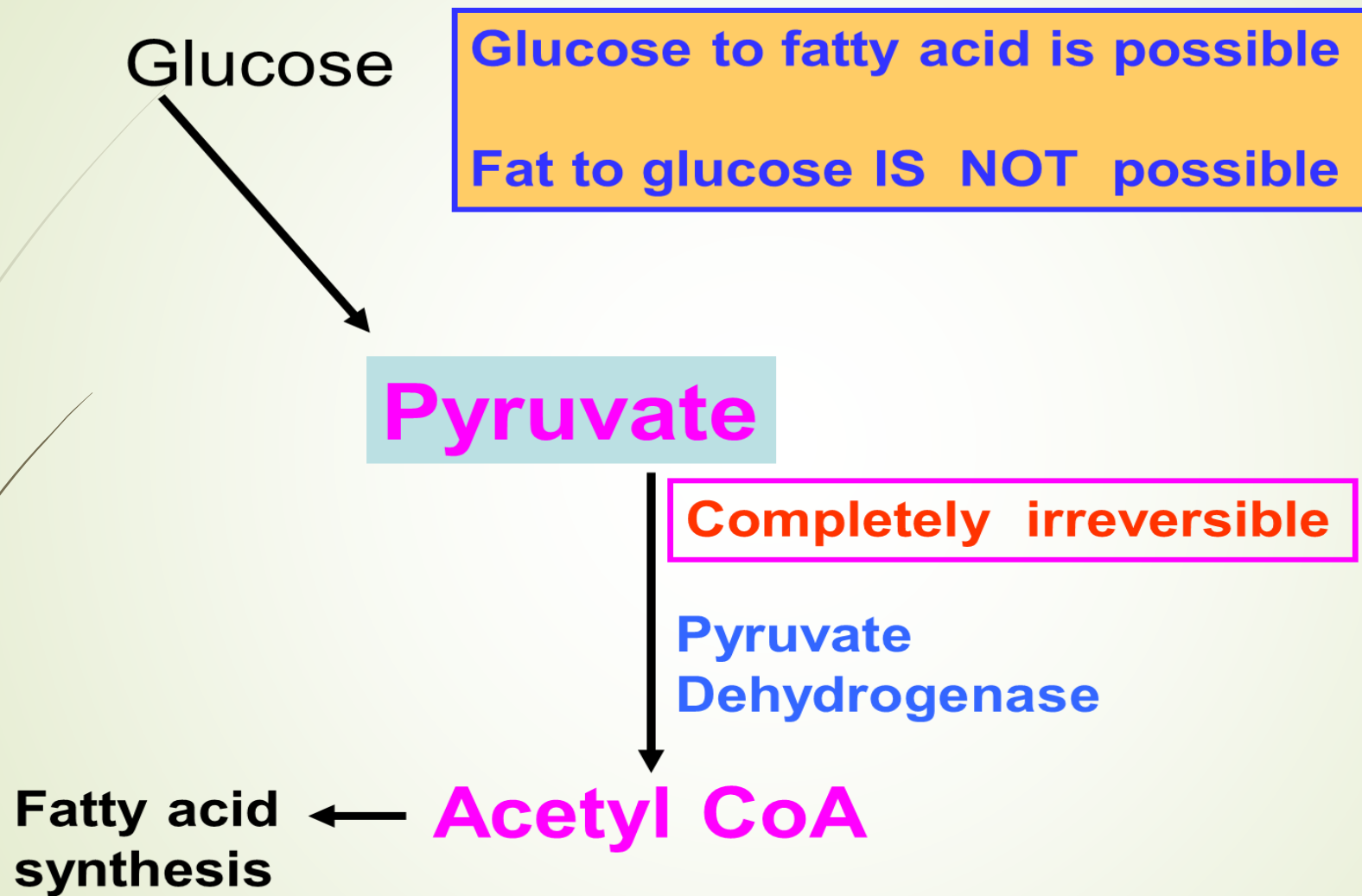
- The 2,3-BPG combines with hemoglobin, and **reduces the affinity towards oxygen**. So, in presence of 2,3-BPG, oxyhemoglobin will unload oxygen more easily in tissues.
- Under **hypoxic conditions the 2,3-BPG concentration in the RBC increases**, thus favoring the release of oxygen to the tissues even when pO_2 is low.
- The compensatory increase in 2,3-BPG in high altitudes favors oxygen dissociation. BPG is increased in fetal circulation.
- In this shunt pathway, **no ATP is generated**

Metabolic Fate of Pyruvate

Under aerobic conditions, pyruvate is converted to acetyl CoA which enters the TCA cycle to be oxidised to CO₂. ATP is generated. Glycolysis is taking place in cytoplasm. So pyruvate is generated in cytoplasm. This is transported into mitochondria by a **pyruvate transporter**.



Pyruvate as a metabolic junction point



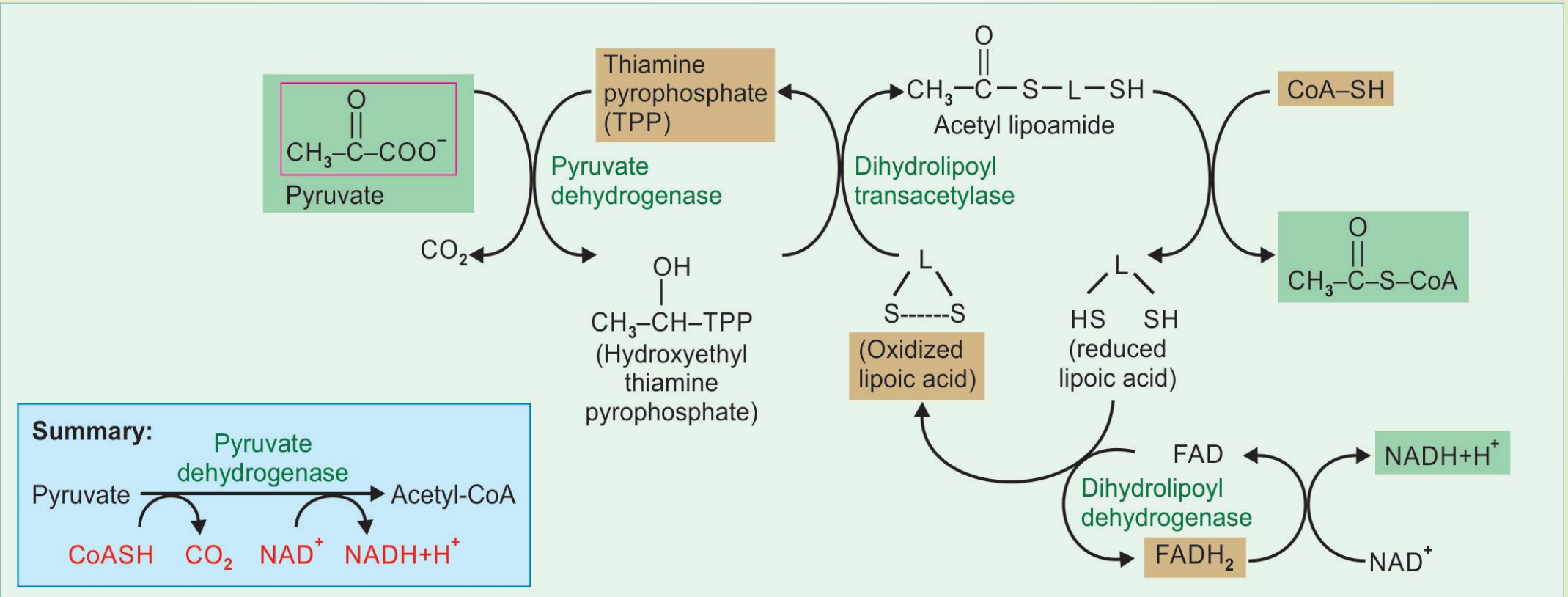
Pyruvate Dehydrogenase Complex (PDH complex)

3 COMPONENTS :

- ✓ Pyruvate decarboxylase / dehydrogenase
- ✓ Dihydrolipoyl transacetylase
- ✓ Dihydro lipoyl dehydrogenase

Co-factors / co-enzymes:

- ✓ Thiamine pyrophosphate
- ✓ lipoamide
- ✓ CoA
- ✓ FAD
- ✓ NAD

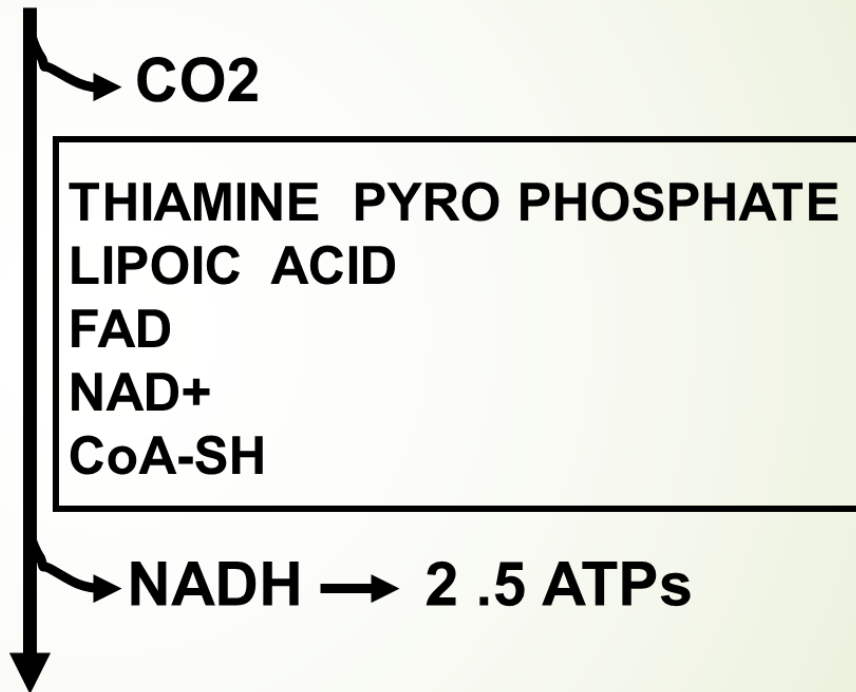


Pyruvate dehydrogenase

Pyruvate
Dehydrogenase
COMMITTED STEP

**Completely
irreversible**

Pyruvate



Acetyl CoA

CH₃—CO--SCoA

Regulation of PDH

- ✓ End product inhibition by
 - acetyl CoA
 - NADH
- ✓ Co-valent modification of PDH enzyme

CLINICAL ASPECTS

1. Thiamine deficiency:
 - PDH activity decreased
 - pyruvate is converted to lactate
 - lactic acidosis
2. Inherited deficiency of glycolytic enzymes
 - Pyruvate kinase
 - aldolase

Clinical Applications of Glycolytic Enzymes

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1. **Lactic acidosis** may be seen in hypoxia, shock, pulmonary failure, alcohol abuse, diabetes mellitus and mitochondrial cytopathies.
2. **Deficiency of glycolytic enzymes:** These conditions are rare, out of which **pyruvate kinase** deficiency and **hexokinase** deficiency are comparatively common. They lead to hemolytic anemia, because energy depleted RBCs are destroyed.
3. **Pyruvate dehydrogenase (PDH):** PDH requires thiamine pyrophosphate (TPP); this explains the serious afflictions in **beriberi** due to thiamine deficiency. TPP deficiency in alcoholism causes pyruvate accumulation in tissues and resultant lactic acidosis. Inherited PDH deficiency may also lead to lactic acidosis.

Gluconeogenesis

Definition

It is the process by which glucose is produced from non-carbohydrate precursors.

Site

Gluconeogenesis occurs mainly in the liver, and to a lesser extent in the renal cortex. The pathway is partly mitochondrial and partly cytoplasmic.

Key Gluconeogenic Enzymes

1. Pyruvate carboxylase
2. Phosphoenol pyruvate carboxykinase
3. Fructose-1-6-bisphosphatase
4. Glucose-6-phosphatase

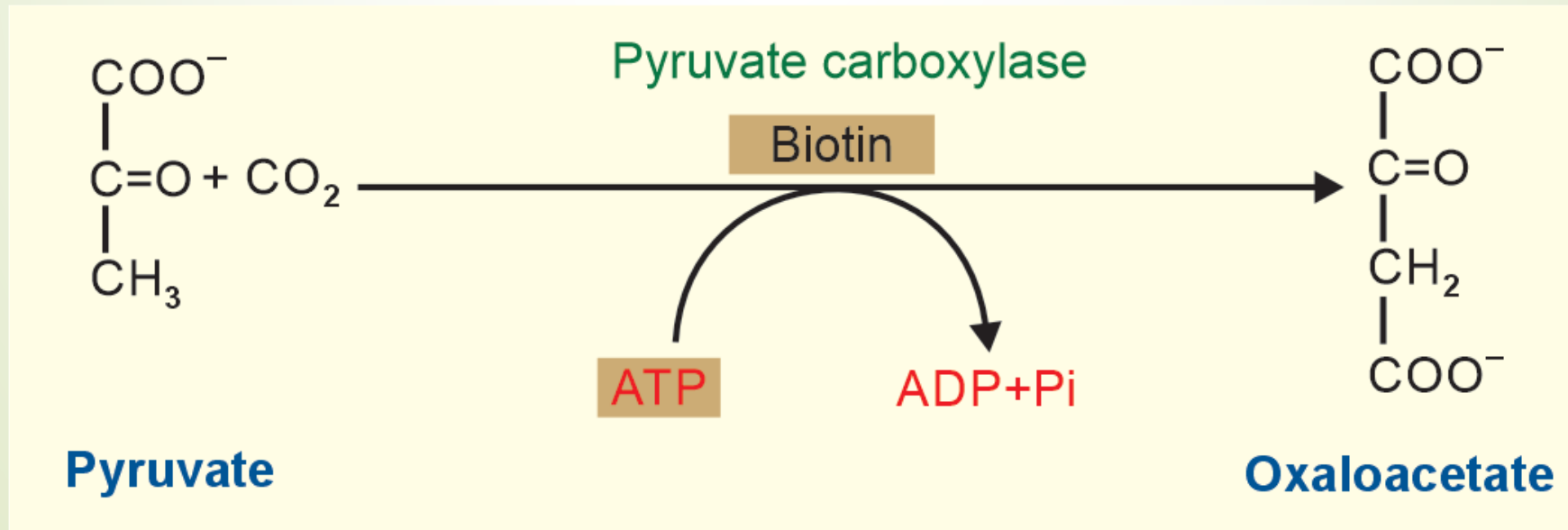
STEPS OF GLUCONEOGENESIS

Step 1: Pyruvate Carboxylase Reaction

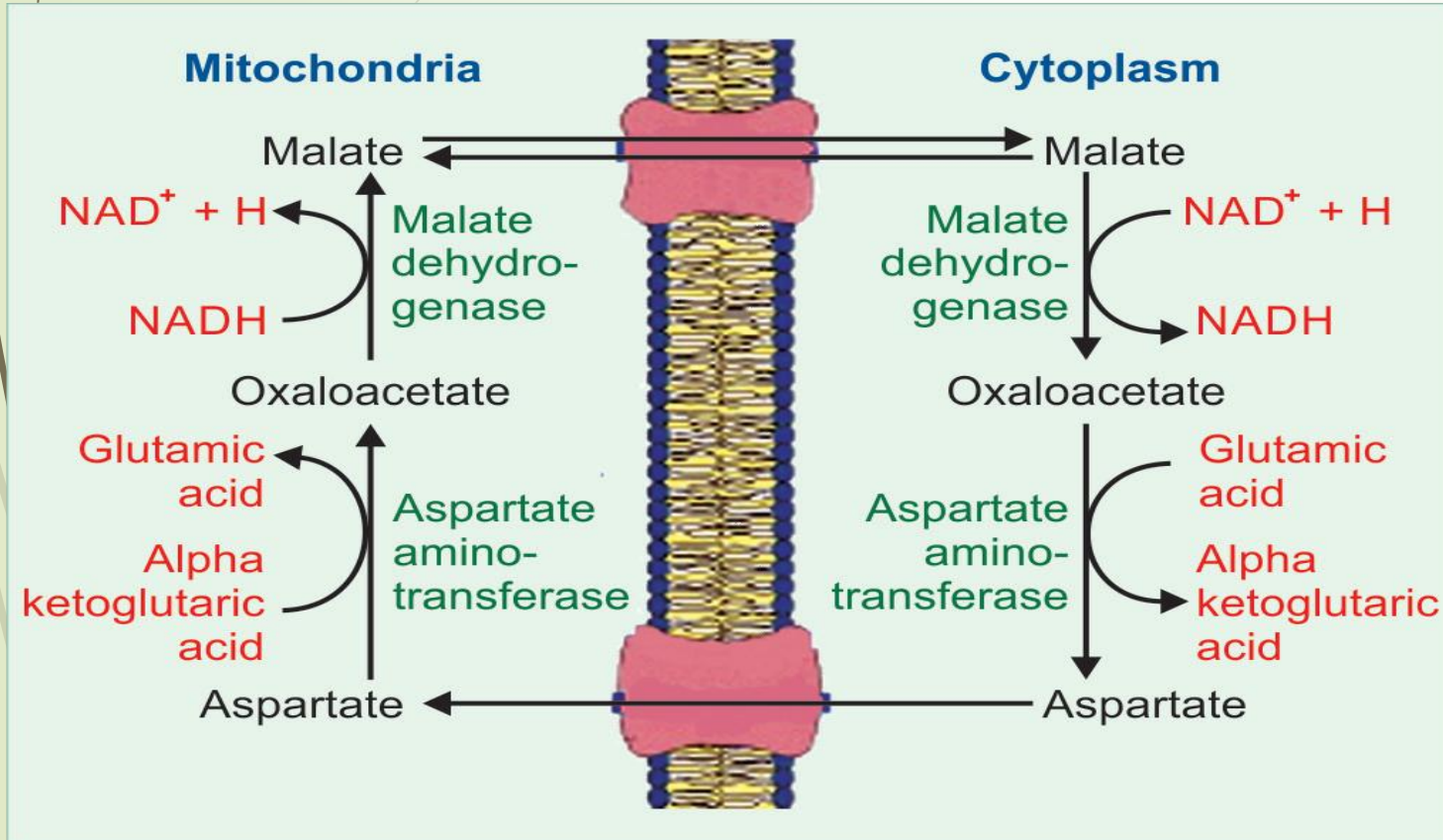
Pyruvate in the cytoplasm enters the mitochondria.

Then, the carboxylation of pyruvate to oxaloacetate is catalyzed by a mitochondrial enzyme, pyruvate carboxylase.

It needs the co-enzymes biotin and ATP.



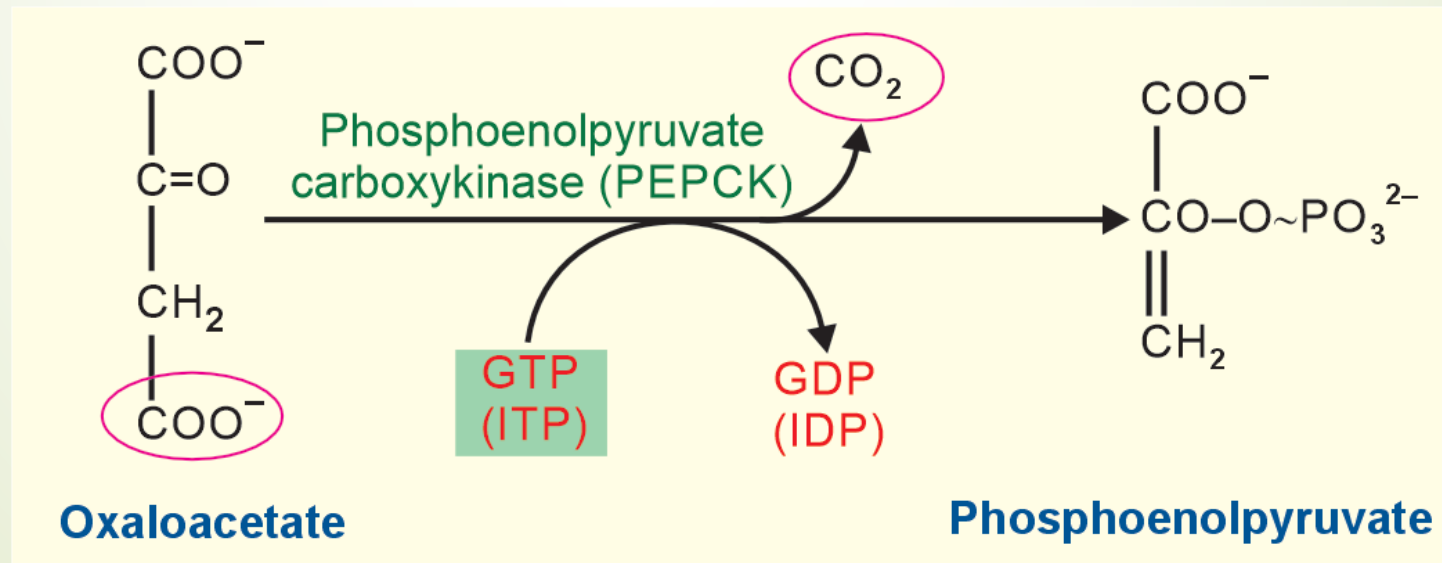
STEP 2: MALATE ASPARTATE SHUTTLE



- ❖ The carboxylation of pyruvate (previous reaction) takes place in mitochondria. So, oxaloacetate is generated inside the mitochondria. This **oxaloacetate has to be transported from mitochondria to cytosol**, because further reactions of gluconeogenesis are taking place in the cytosol. This is achieved by the **malate aspartate shuttle**.
- ❖ Oxaloacetate is first converted to malate, which traverses the membrane and reaches the cytoplasm. Malate is then re-converted to oxaloacetate. Malate dehydrogenase is present in both mitochondria and cytoplasm.

STEP 3: PHOSPHOENOLPYRUVATE CARBOXYKINASE

- ❖ In the cytoplasm, phosphoenolpyruvate carboxykinase (PEPCK) enzyme then converts oxaloacetate to phosphoenolpyruvate by removing a molecule of CO₂.
- ❖ GTP or ITP donates the phosphate. The net effect of these two reactions is the conversion of pyruvate to phosphoenolpyruvate.





STEPS 4, 5, 6, 7, 8: PARTIAL REVERSAL OF GLYCOLYSIS

- ❖ The phosphoenolpyruvate undergoes further reactions catalyzed by the glycolytic enzymes to form fructose-1,6- biphosphate (see glycolysis steps 8, 7, 6, 5 and 4). All these reactions are freely reversible.

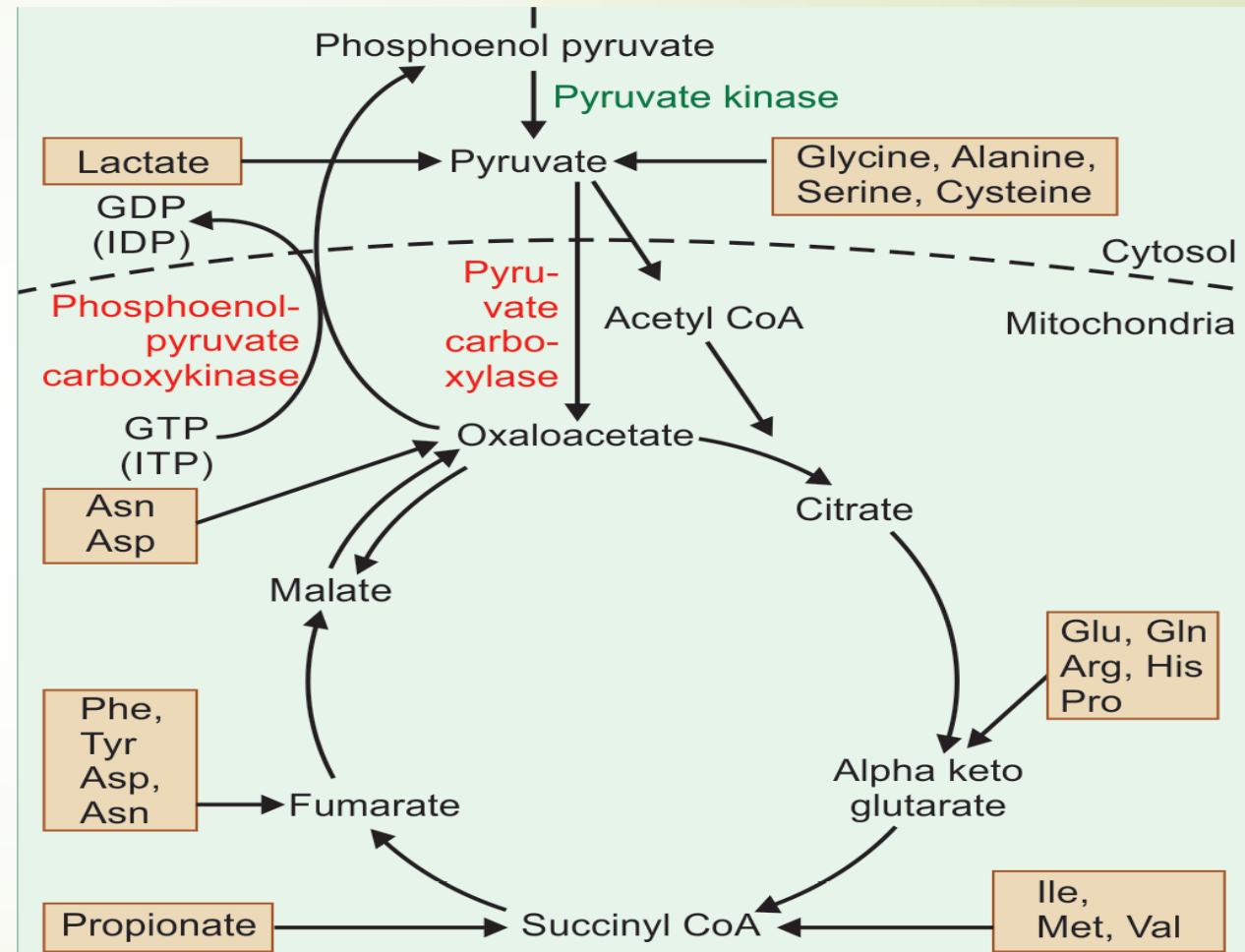
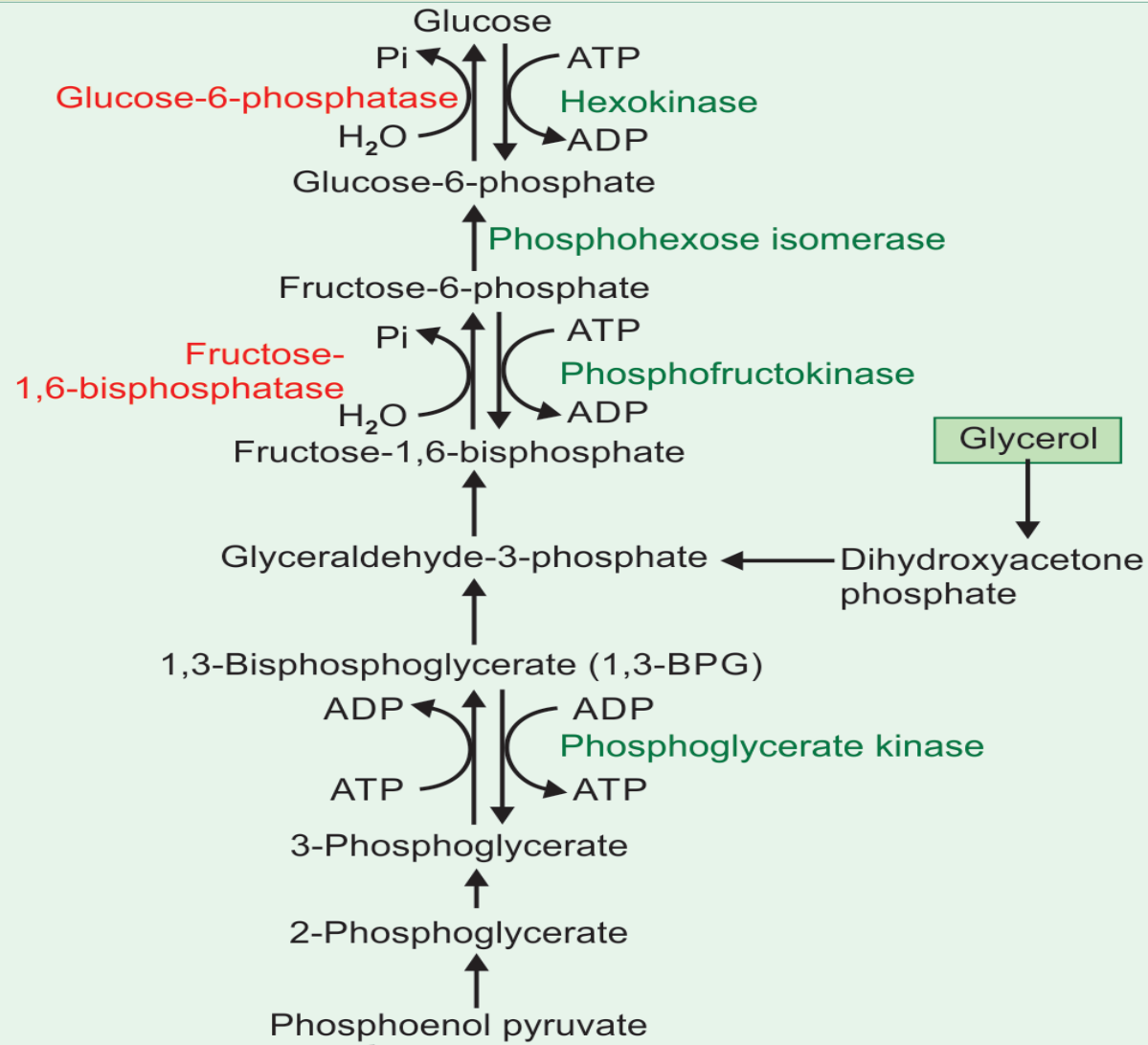
STEP 9: FRUCTOSE-1, 6-BISPHOSPHATASE

- ❖ Fructose-1,6-bisphosphate is then acted upon by fructose- 1,6-bisphosphatase to form fructose-6-phosphate. This will bypass the step of the PFK reaction (see step 3 of glycolysis).
- ❖ Fructose-1-6- Fructose-1,6-bisphosphatase bisphosphate
- ❖ Fructose-6- phosphate + Pi
- ❖ Then fructose-6-phosphate is isomerized to glucose-6- phosphate by the freely reversible reaction catalyzed by hexose phosphate isomerase (step 2 in glycolysis).

STEP 10: GLUCOSE-6-PHOSPHATASE

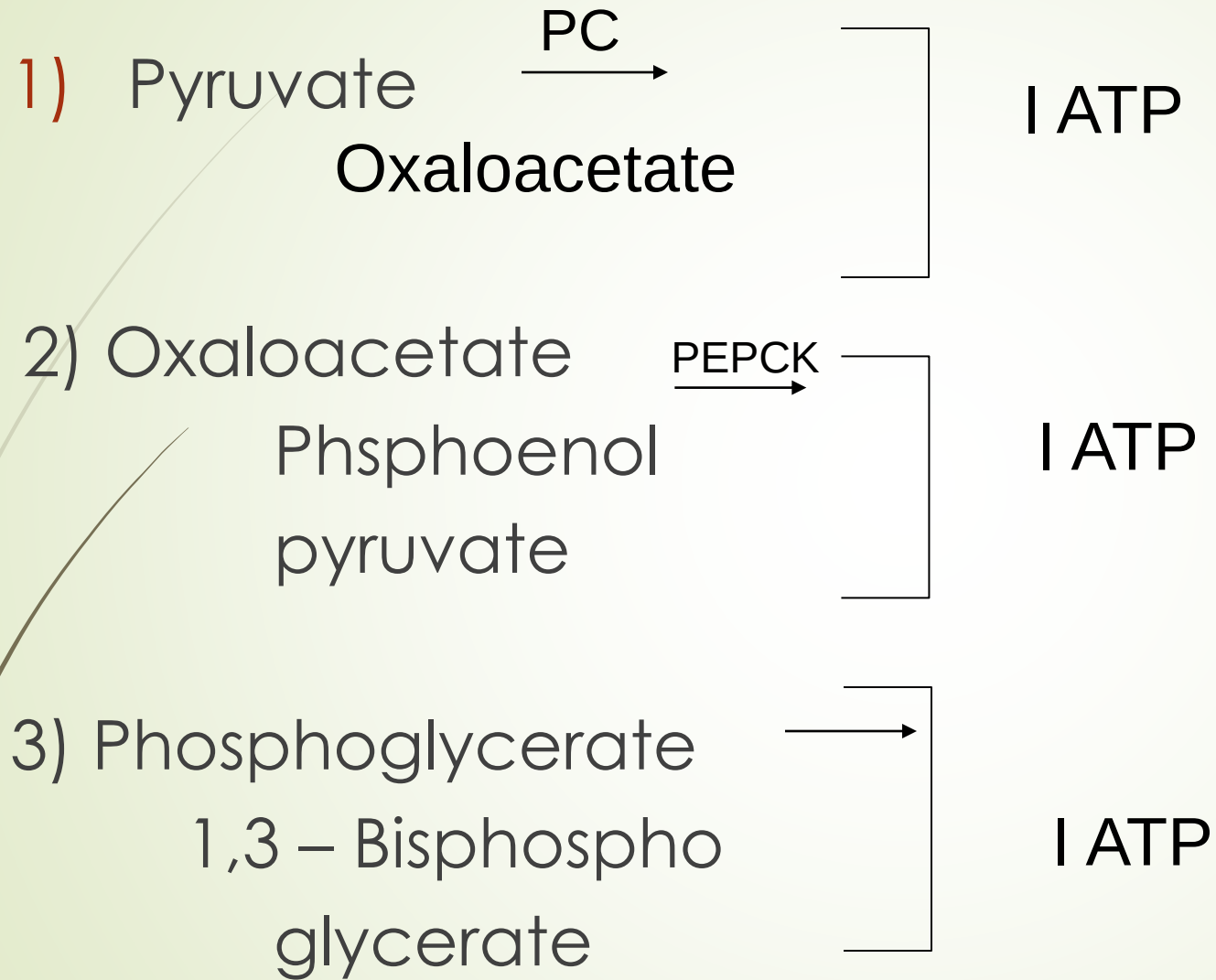
- ❖ The glucose-6-phosphate is hydrolyzed to free glucose by glucose-6-phosphatase.
- ❖ $\text{Glucose-6-phosphate} + \text{H}_2\text{O} \rightarrow \text{Glucose} + \text{P}_i$
- ❖ Glucose-6-phosphatase is **active in the liver**. It is present in kidney and intestinal mucosa to a lesser extent but is **absent in muscle**.

Irreversible steps in glycolysis	Corresponding key gluconeogenic enzymes
Pyruvate kinase (Step 9)	Pyruvate carboxylase; Phosphoenol pyruvate carboxy kinase
Phosphofructokinase (Step 3)	Fructose-1,6- bisphosphatase
Hexokinase (Step 1)	Glucose-6- phosphatase



Key gluconeogenic enzymes are shown in red
 Key glycolytic enzymes are depicted in green
 Substrates for gluconeogenesis are shown inside brown boxes

Gluconeogenesis - Energy Requirement



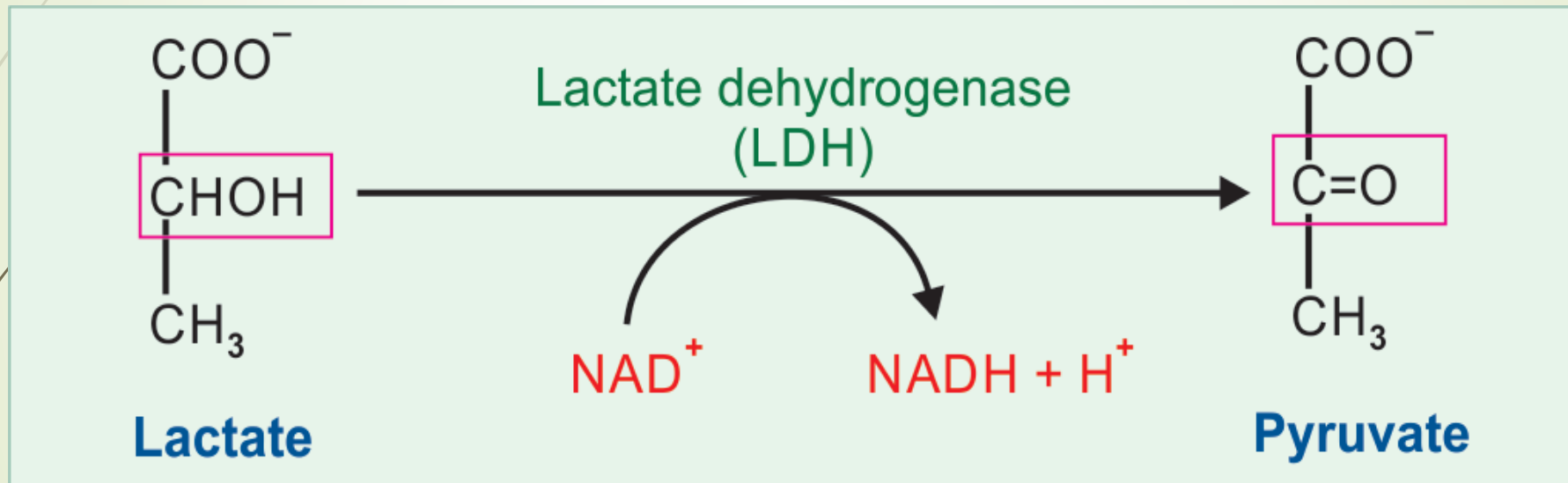
3 x 2 = 6 ATP molecules for 1 molecule of glucose

NON-CARBOHYDRATE PRECURSORS

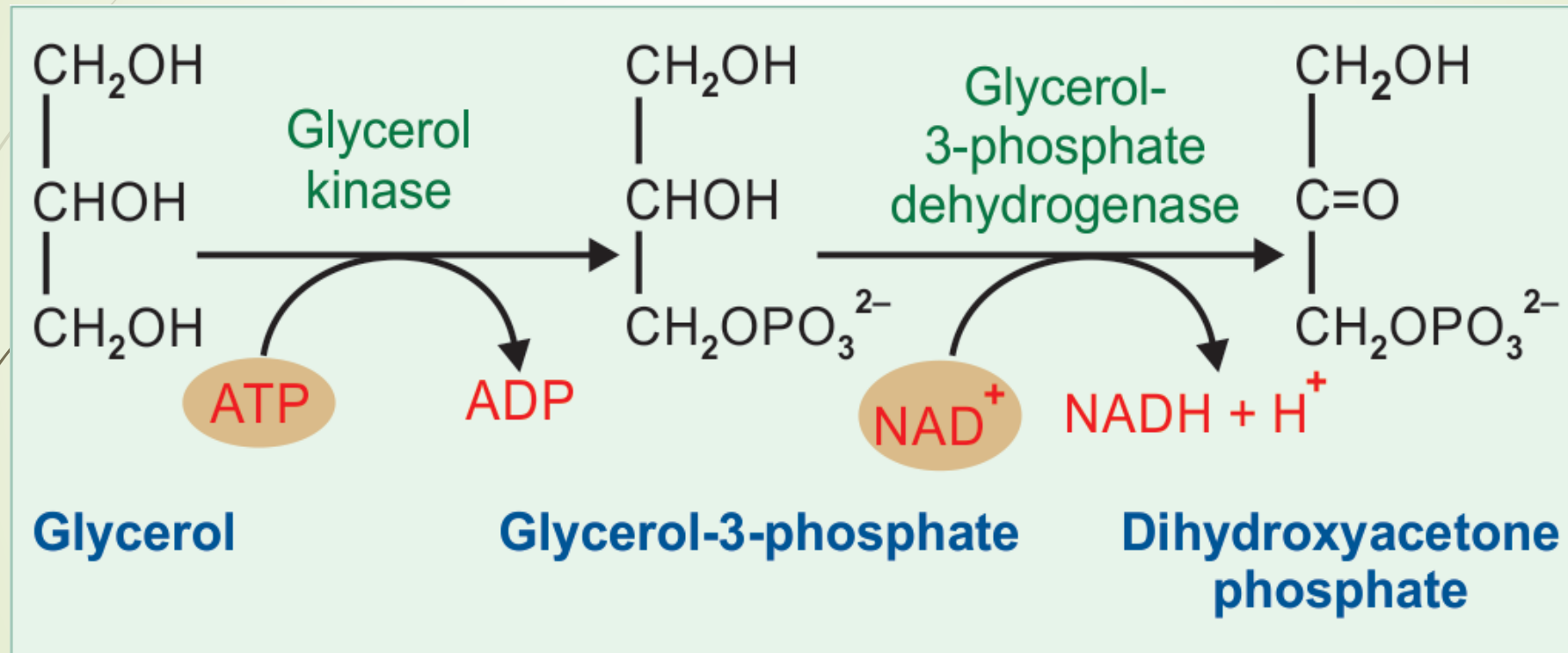
- LACTATE
- GLUCOGENIC AMINO ACIDS
- GLYCEROL
- PROPIONIC ACID

~~FATTY ACID~~ cannot give rise to glucose

Gluconeogenesis Substrate Lactate



Gluconeogenesis from Glycerol

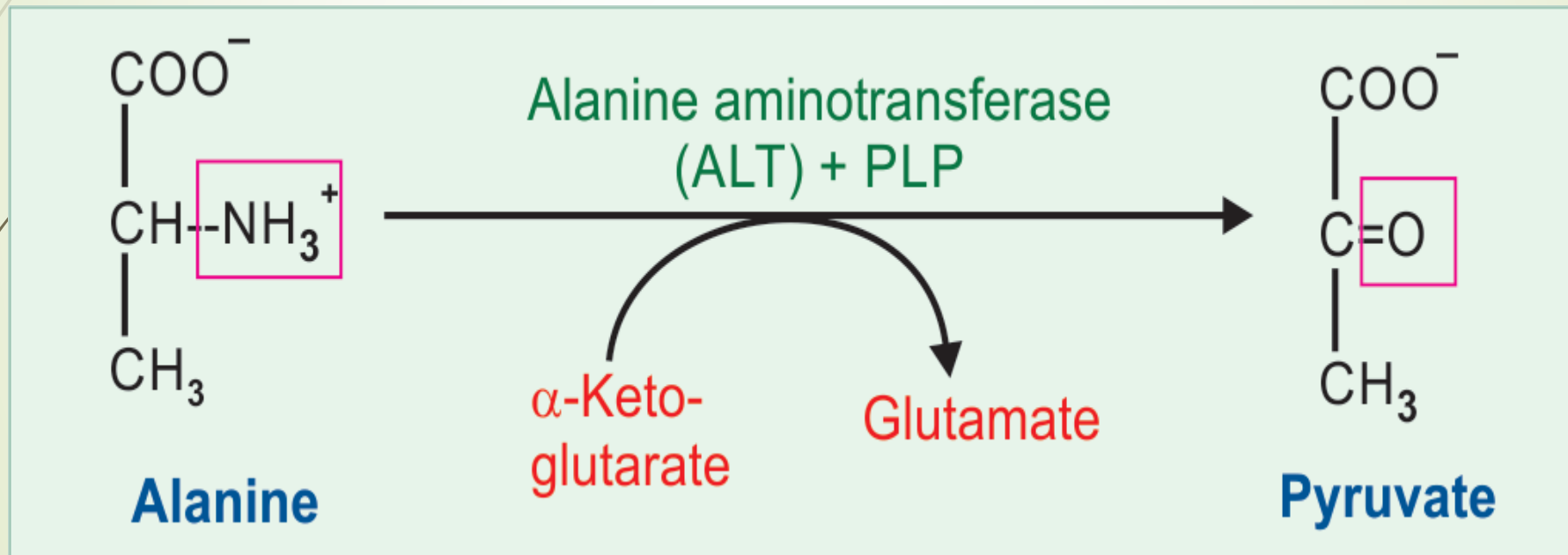


Transamination Reaction

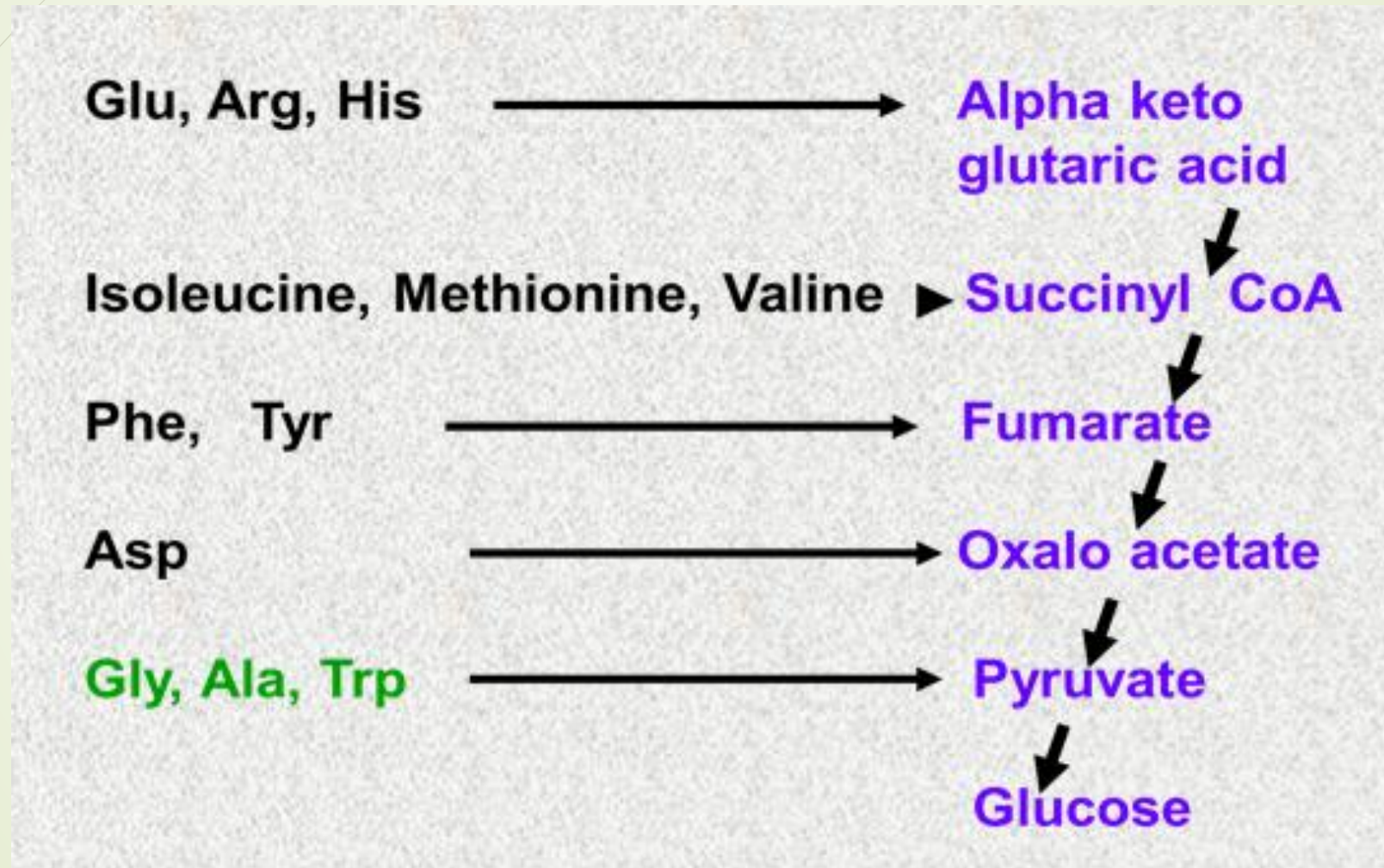
Alanine

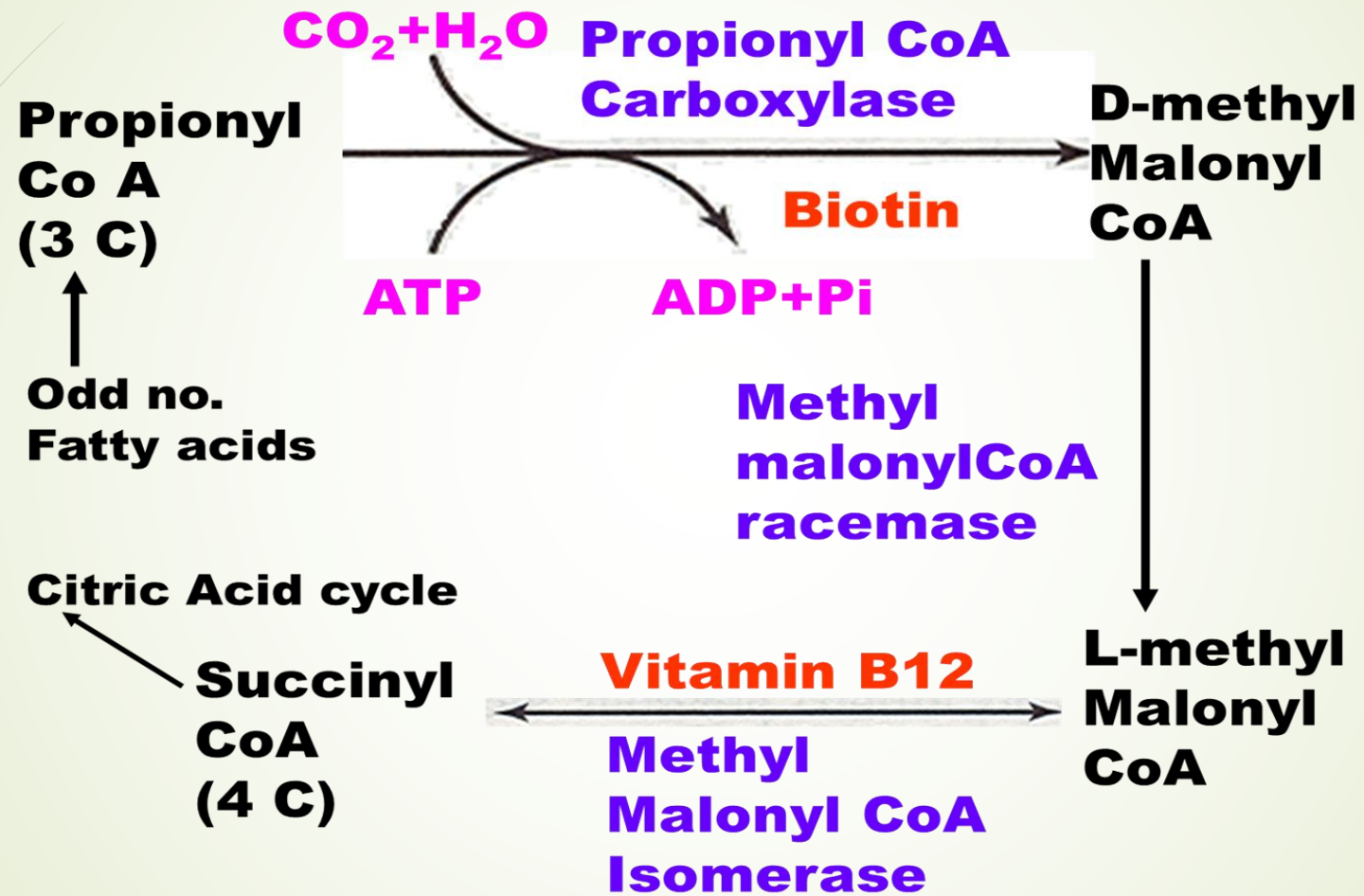


Pyruvate

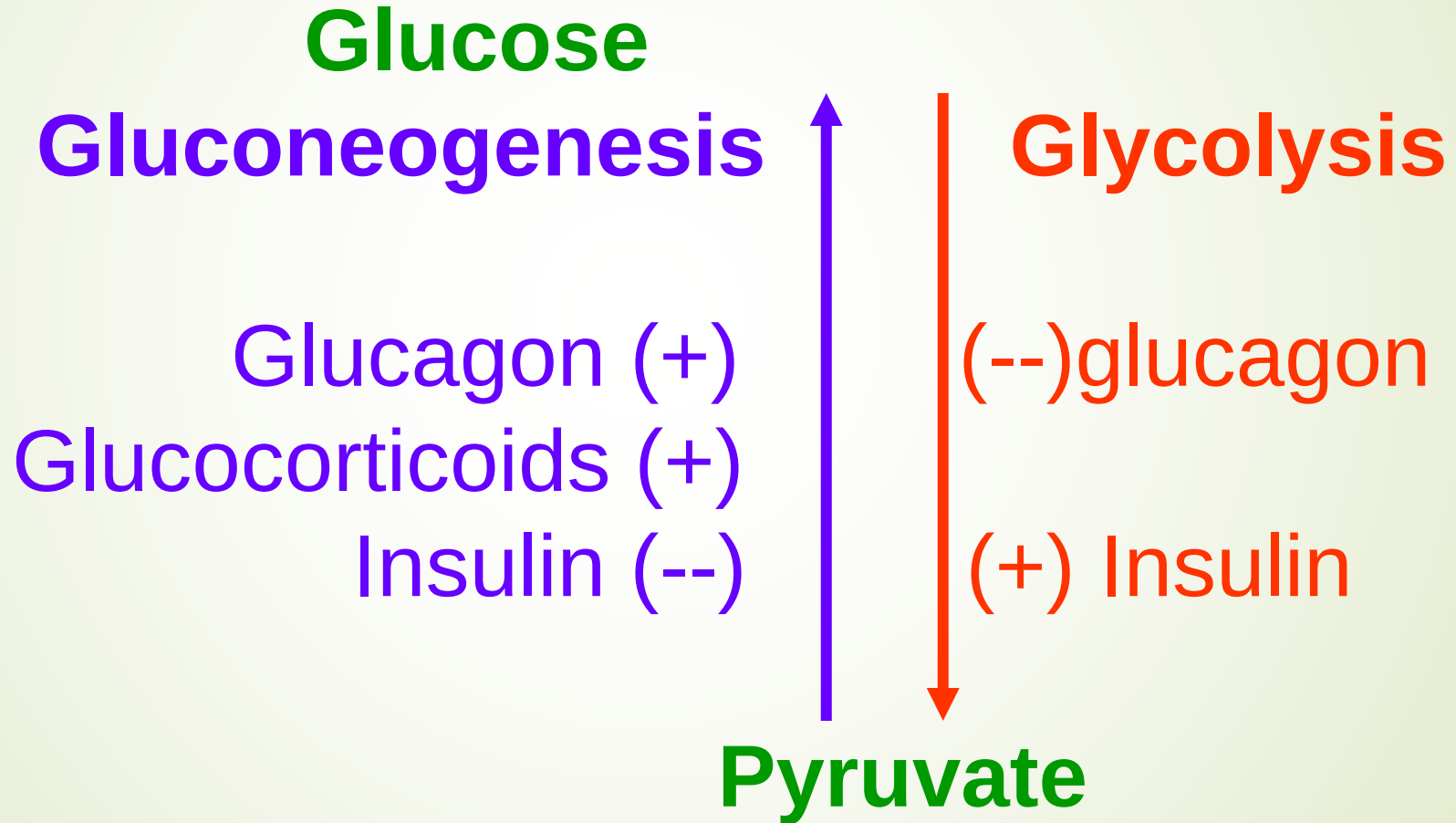


Gluconeogenesis from Glucogenic Amino Acids

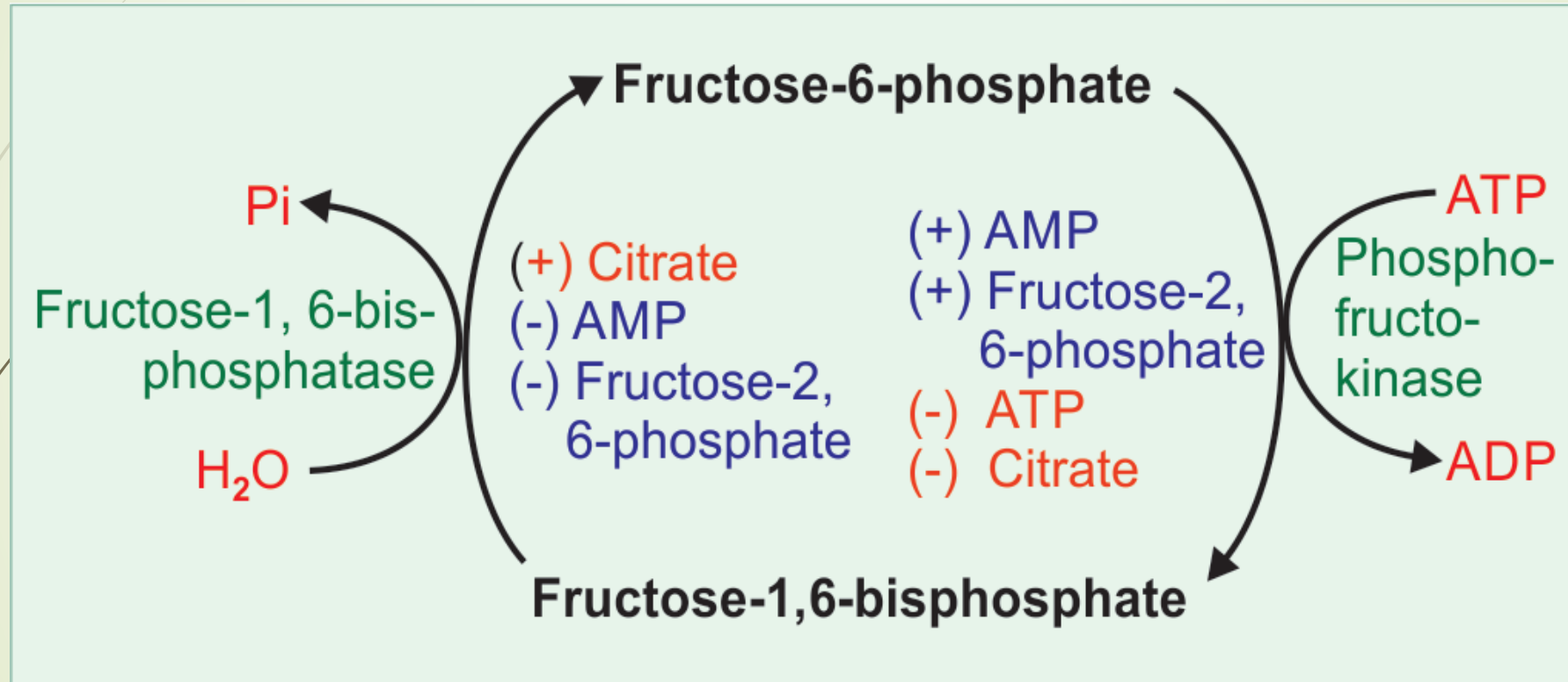


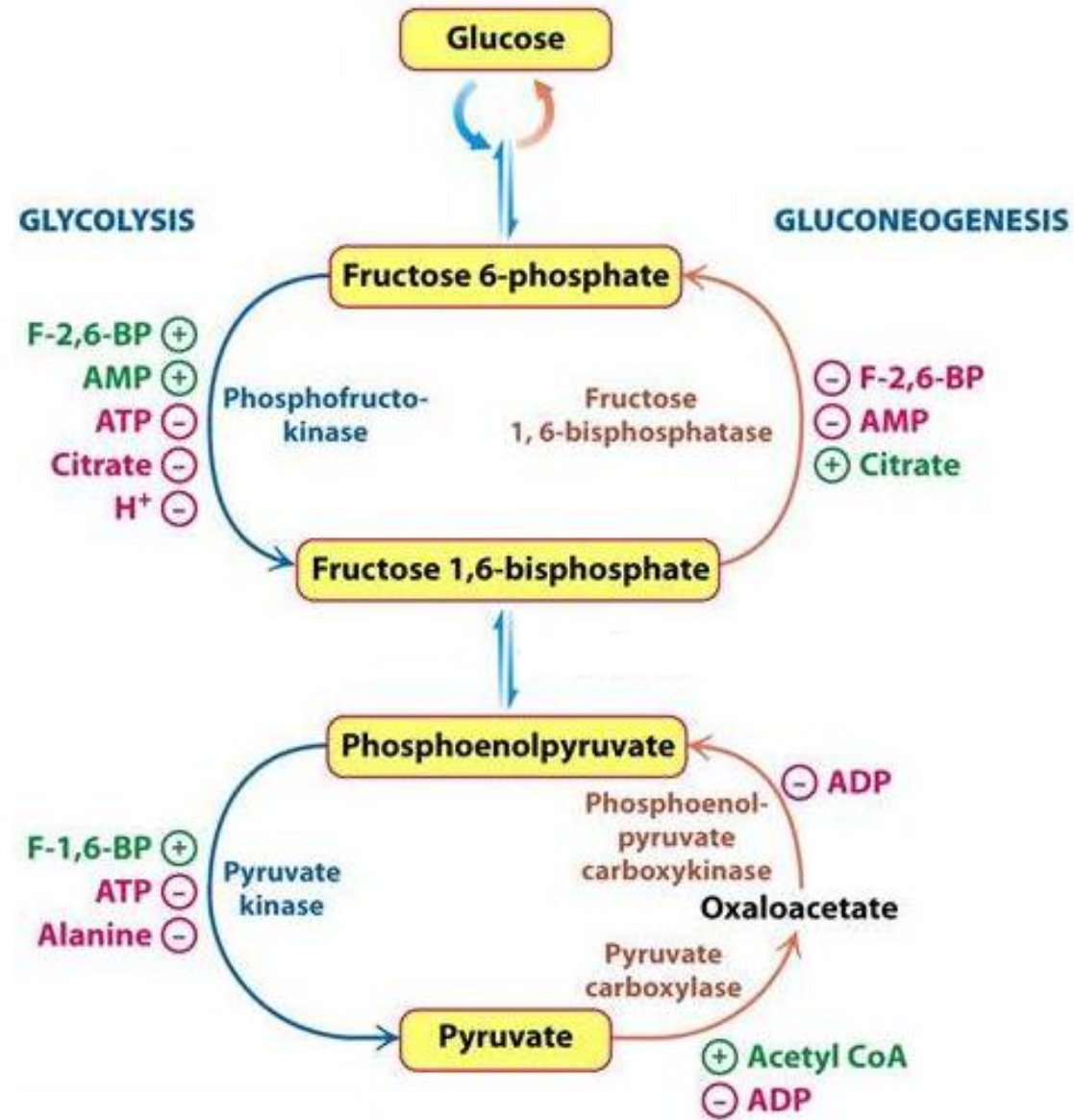


Hormonal Regulation of Gluconeogenesis



Reciprocal Regulation of Enzymes





Physiological Significance of Gluconeogenesis

Maintenance of blood glucose level especially under conditions of starvation.

The brain has a minimum need of 120 grams of glucose per day.

The body stores of glycogen are depleted within the first 12-18 hours of fasting.

On prolonged starvation

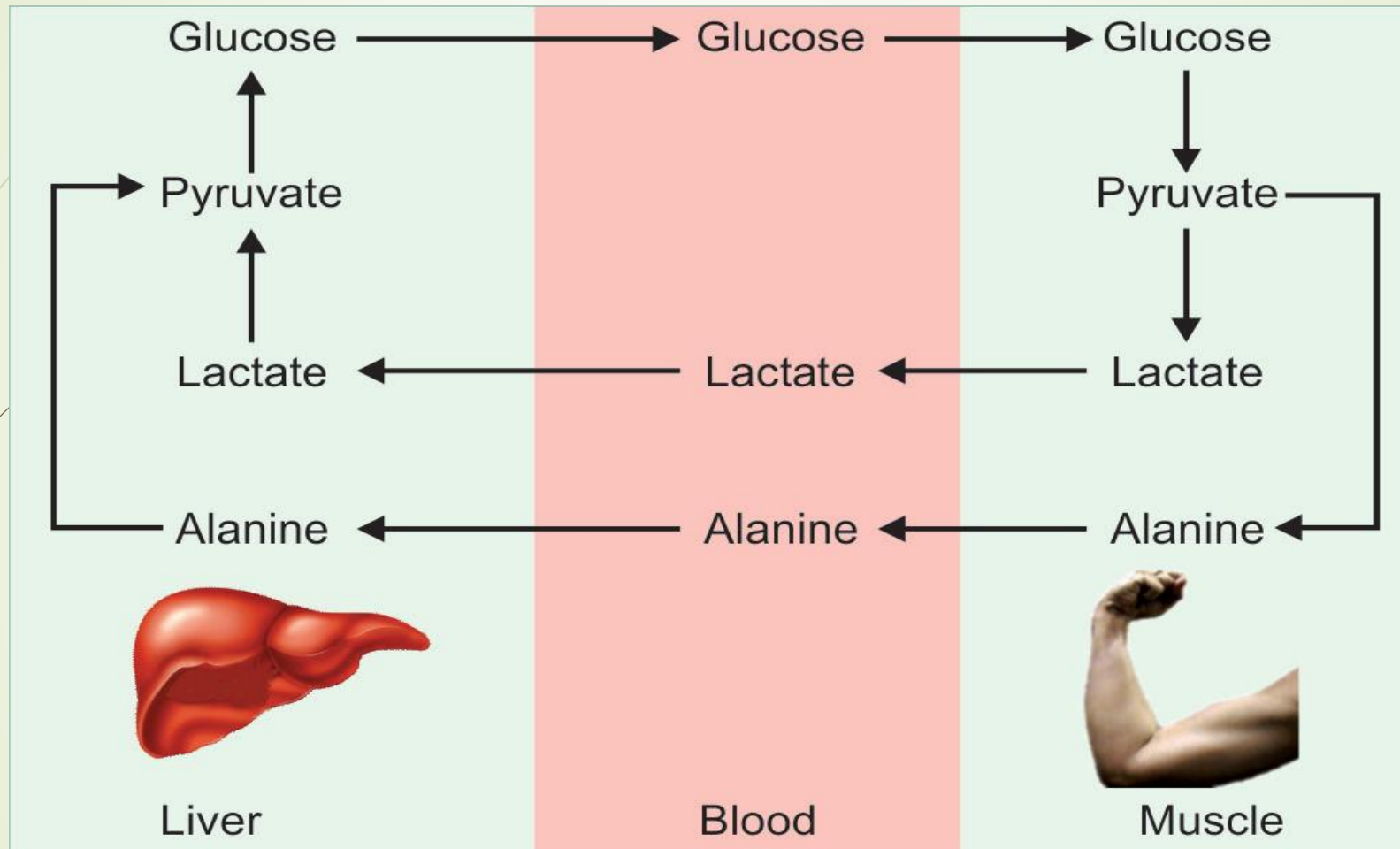
gluconeogenesis is speeded up

Protein catabolism provides the substrates, glucogenic amino acids for gluconeogenesis

Clinical Significance of Gluconeogenesis

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1. **Pyruvate carboxylase deficiency:** It is a rare inborn error of metabolism manifested with neurological symptoms and mental retardation, due to deficient fuel supply to the nervous system. Accumulation of lactic acid and keto acids leads to metabolic acidosis. Hyperammonemia is also seen. It is an autosomal recessive condition
2. **Malignant hyperthermia:** This may occur when halothane is given as an anesthetic to certain persons. There is inappropriate release of calcium from sarcoplasmic reticulum. This results in uncontrolled heat generation, damage of muscle cells
3. **Ethanol (Ethyl alcohol):** It inhibits gluconeogenesis. During the metabolism of ethanol the level of cytoplasmic NADH is raised. Thus, the Pyruvate $\rightarrow$ Malate $\rightarrow$ Oxaloacetate reactions are reversed. So, excessive ingestion of alcohol results in hypoglycemia. Lactate also accumulates as NADH level is high.



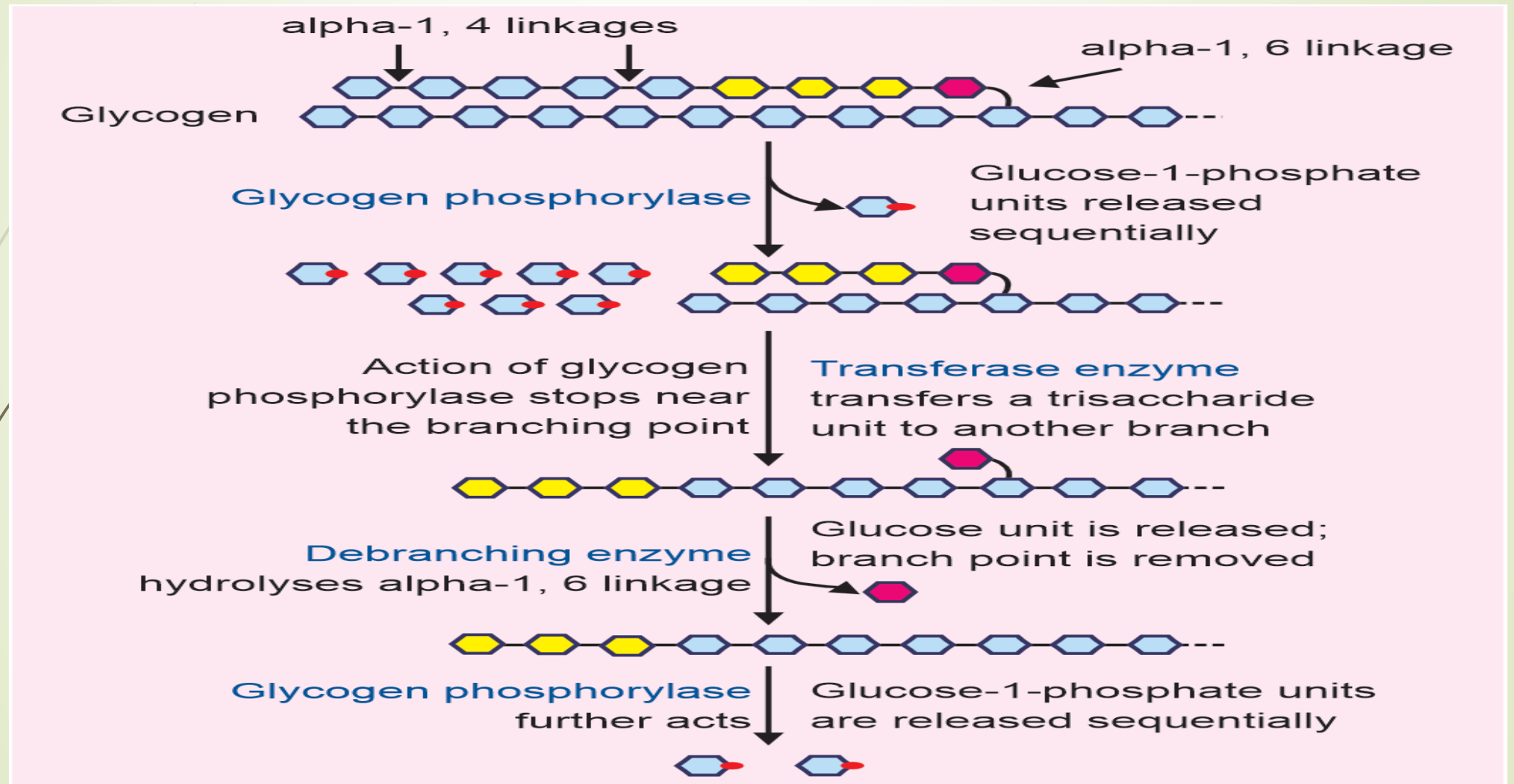
Cori's cycle (upper circle) and Glucose-alanine cycle (lower circle)

GLYCOGEN METABOLISM

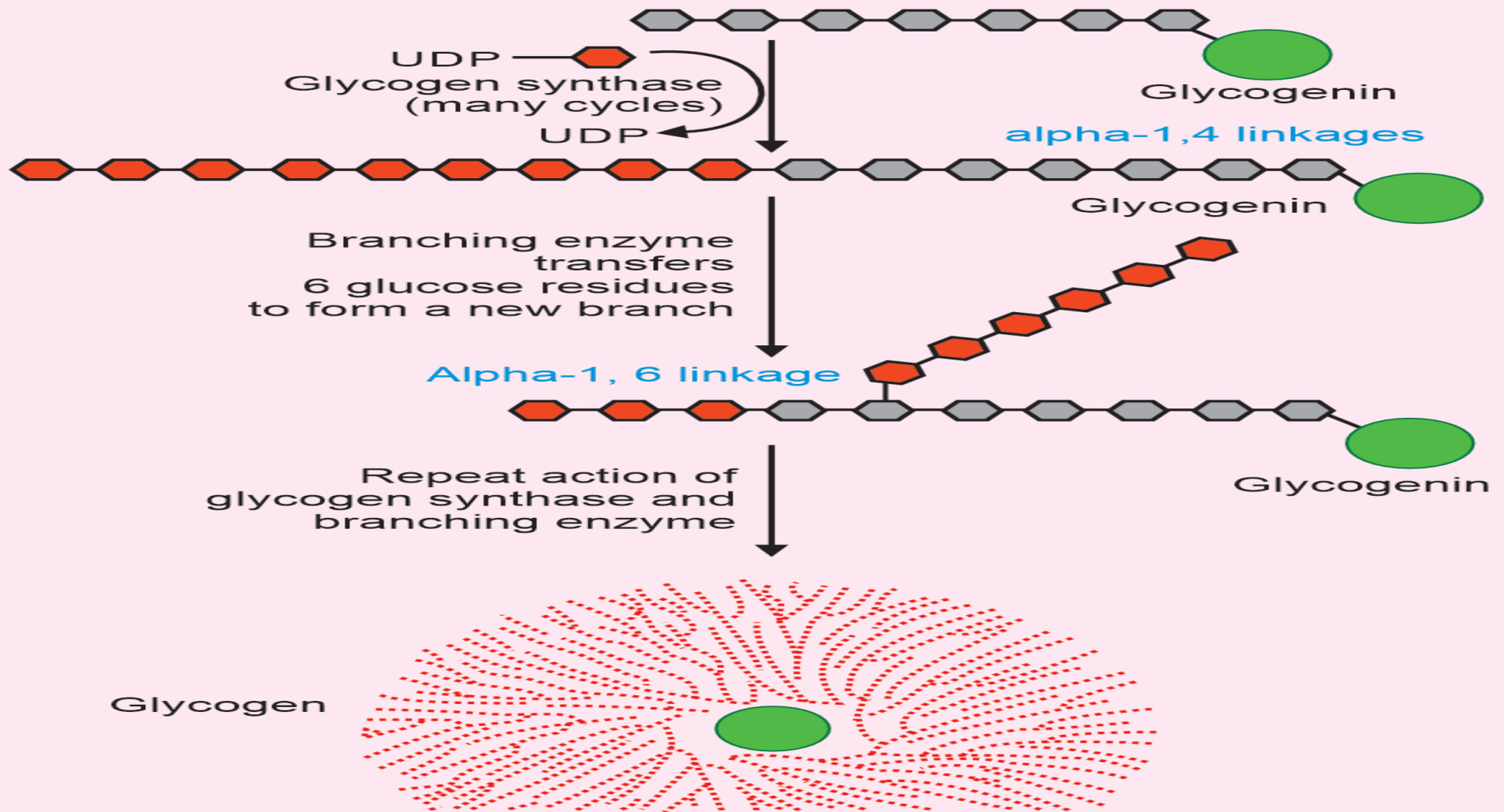
GLYCOGEN METABOLISM

- ❖ Glycogen is the storage form of carbohydrates in the human body. The major sites of storage are the liver and muscle. The major function of liver glycogen is to provide glucose during fasting.
- ❖ The glycogen content of the liver (10 g/100 g tissue) is more than in the skeletal muscle (1–2 g/100 g). But the total quantity of muscle glycogen is more than liver glycogen because of the larger muscle mass. Glycogen is also found in the brain and kidneys.
- ❖ When blood glucose level falls, liver glycogen is broken down and helps to maintain blood glucose level. After taking food, blood glucose tends to rise, which causes glycogen deposition in the liver.
- ❖ About 5 hours after taking food, the blood glucose tends to fall. But, glycogen is broken down to glucose so that the energy needs are met.

GLYCOGEN BREAKDOWN (GLYCOGENOLYSIS) PATHWAY

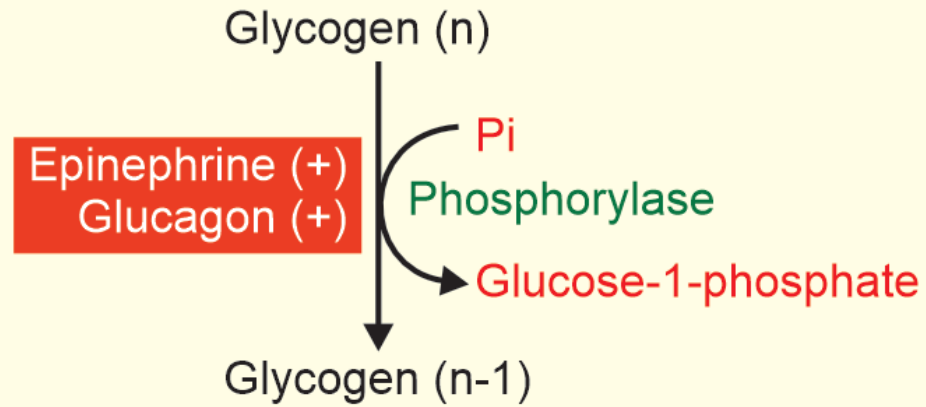


GLYCOGEN SYNTHESIS PATHWAY

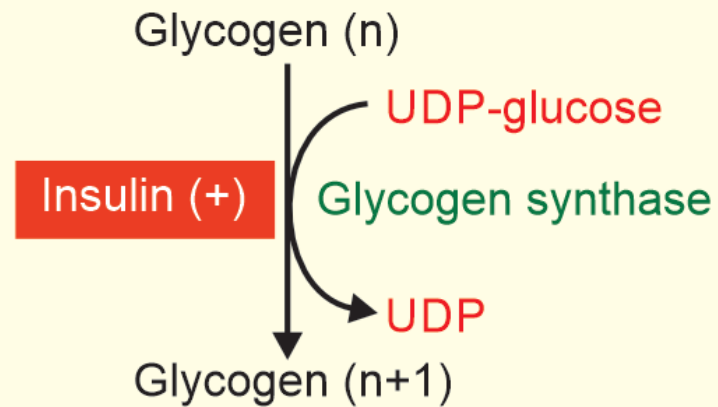


EFFECTS OF HORMONES ON GLYCOGEN

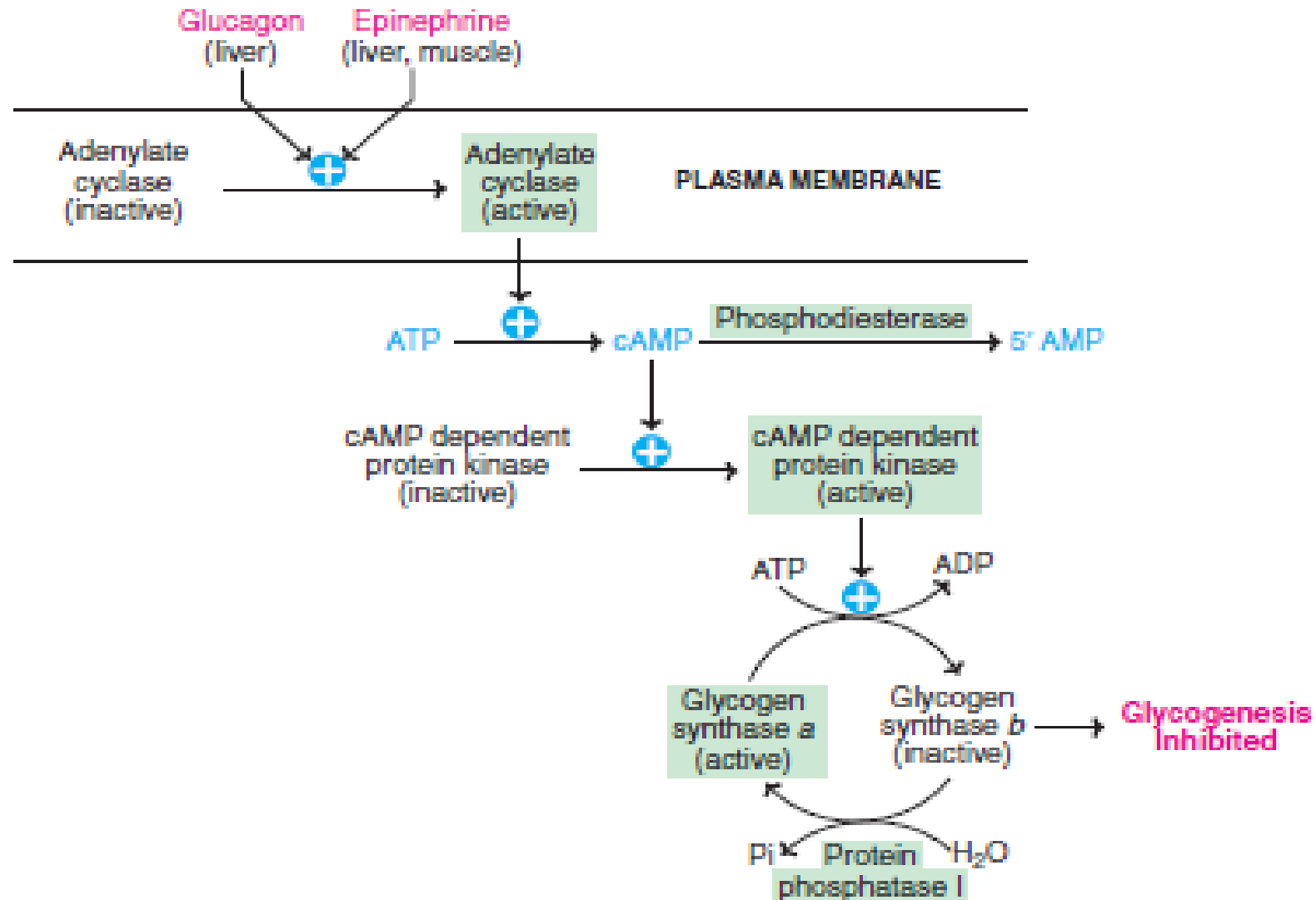
Glycogenolysis



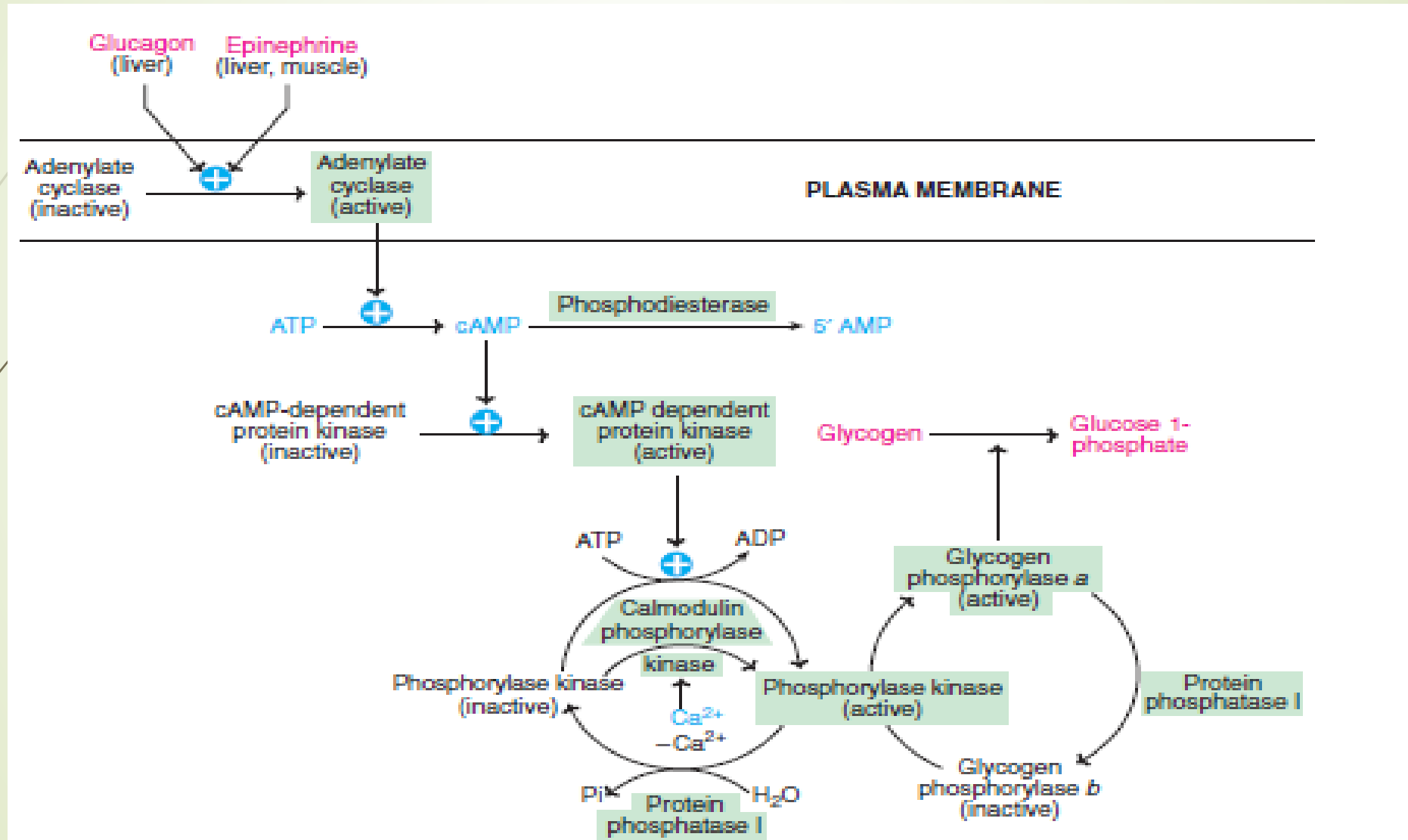
Glycogen synthesis



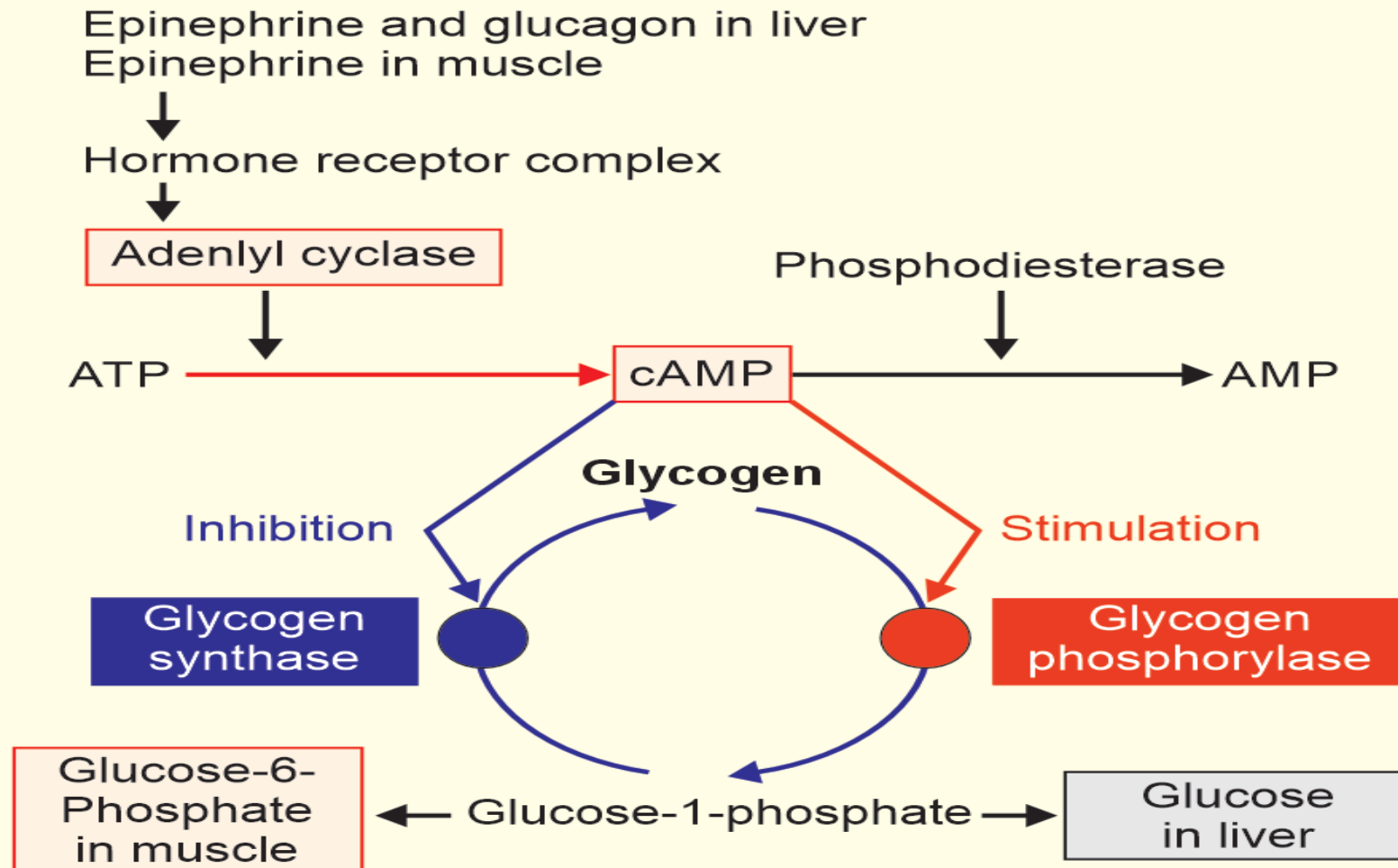
HORMONAL REGULATION OF GLYCOGEN SYNTHESIS



HORMONAL REGULATION OF GLYCOGEN BREAKDOWN



RECIPROCAL REGULATION OF GLYCOGENOLYSIS AND GLYCOGEN SYNTHESIS BY CYCLIC AMP



GLYCOGEN STORAGE DISEASES

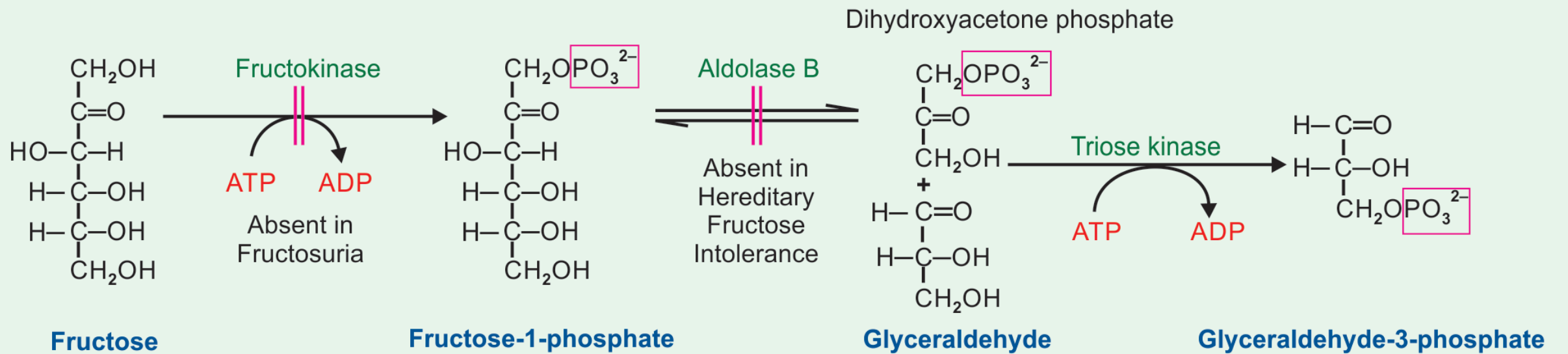
Table 8.12: Glycogen storage diseases.

Type	Name	Deficient enzyme	Clinical features
Type Ia	von Gierke's disease	Glucose-6-phosphatase	Fasting hypoglycemia; hepatomegaly
Type Ib		Endoplasmic reticulum glucose-6-P transporter	Same as above, plus neutropenia and recurrent infections
Type II	Pompe's disease	Lysosomal alpha glycosidase (GAA)	♦ Accumulation of glycogen in lysosomes of liver, heart and muscle; death before 2 years. May manifest as a congenital type or late onset type. Infants have hypotonia and cardiac insufficiency ♦ Intravenous recombinant alpha-glucosidase has been tried with some success
Type III	Limit dextrinosis/ Cori's disease	Debranching enzyme	Highly branched dextrin accumulates. Fasting hypoglycemia; hepatomegaly
Type IV	Amylopectinosis/ Andersen's disease	Branching enzyme	Glycogen with few branches; hepatosplenomegaly; mild hypoglycemia; death by age of 5. Severe cases may present in with intrauterine hydrops and after birth with severe hypotonia and cardiomyopathy. Varying neuromuscular symptoms like isolated myopathy or upper or lower motor neuron involvement. Muscle biopsy shows an accumulation of abnormal glycogen
Type V	McArdle's disease	Muscle phosphorylase	Exercise intolerance; accumulation of glycogen in muscles; normoglycemia and elevated CK
Type VI	Hers' disease	Liver phosphorylase	Mild hypoglycemia; hepatomegaly; better prognosis than other types
Type VII	Tarui's disease	Muscle PFK	Glycogen in muscles accumulated; exercise intolerance; hemolytic anemia; normoglycemia
Type VIII		Liver phosphorylase kinase	Mild hypoglycemia; better prognosis
Type IX		Muscle phosphorylase kinase	Mild exercise intolerance; better prognosis
Type X		Protein kinase A	Hepatomegaly

Type	Name	Enzyme defect	Organ(s) involved	Characteristic features
I	von Gierke's disease (type I glycogenosis)	Glucose 6-phosphatase	Liver, kidney and intestine	Glycogen accumulates in hepatocytes and renal cells, enlarged liver and kidney, fasting hypoglycemia, lactic acidemia; hyperlipidemia; ketosis; gouty arthritis.
II	Pompe's disease	Lysosomal α -1,4 glucosidase (acid maltase)	All organs	Glycogen accumulates in lysosomes in almost all the tissues; heart is mostly involved; enlarged liver and heart, nervous system is also affected; death occurs at an early age due to heart failure.
III	Cori's disease (limit dextrinosis, Forbe's disease)	Amylo α -1,6-glucosidase (debranching enzyme)	Liver, muscle, heart, leucocytes	Branched chain glycogen accumulates; liver enlarged; clinical manifestations are similar but milder compared to von Gierke's disease.
IV	Anderson's disease (amylopectinosis)	Glucosyl 4-6 transferase (branching enzyme)	Most tissues	A rare disease, glycogen with only few branches accumulate; cirrhosis of liver, impairment in liver function.
V	McArdle's disease (type V glycogenosis)	Muscle glycogen phosphorylase	Skeletal muscle	Muscle glycogen stores very high, not available during exercise; subjects cannot perform strenuous exercise; suffer from muscle cramps; blood lactate and pyruvate do not increase after exercise; muscles may get damaged due to inadequate energy supply.
VI	Her's disease	Liver glycogen phosphorylase	Liver	Liver enlarged; liver glycogen cannot form glucose; mild hypoglycemia and ketosis seen.
VII	Tarui's disease	Phosphofructokinase	Skeletal muscle, erythrocytes	Muscle cramps due to exercise; blood lactate not elevated; hemolysis occurs.

FRUCTOSE METABOLISM

- Fructose is a ketohexose present in fruits, honey and sucrose.
- Fructose is promptly metabolized by the liver.



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- Free fructose is seen in large quantities in **seminal plasma**.
 - **The energy for mobility of spermatozoa** is mainly derived from fructose.
 - Fructose is secreted by seminal vesicles. In some persons azoospermia is seen due to a block in the duct.
 - In such persons fructose is estimated in semen.
 - If fructose is present, the block is above the seminal vesicular duct; if absent, block is after the seminal vesicles.

Hereditary Fructose Intolerance (HFI)

Autosomal recessive

Inborn error of metabolism

1 in 70 persons are carriers of abnormal gene

The defect is in Aldolase-B

Manifested by vomiting and loss of appetite.

Hepatomegaly and jaundice may occur.

When sucrose or fructose is introduced in diet of infants usually around 6 months.

**Fructose is excreted in urine ;
urine gives positive Benedict's test.**



Accumulation of fructose-1-phosphate will inhibit glycogen phosphorylase.

This leads to accumulation of glycogen in liver and associated hypoglycemia

The infants often fail to thrive.

If liver damage progresses,

Cirrhosis, Death.

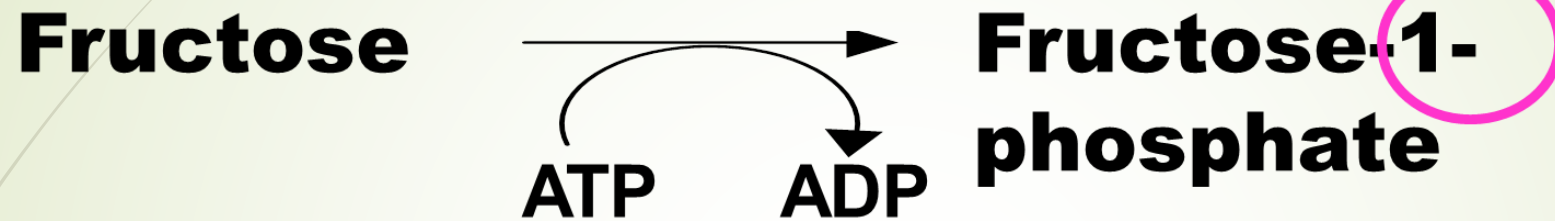
Withdrawal of fructose from the diet will immediately relieve the symptoms.

- 
- 
- This is seen when sucrose (containing fructose) is introduced in the diet of infants, usually around 6 months of age.
 - Accumulation of fructose-1-phosphate will inhibit glycogen phosphorylase.
 - This leads to accumulation of glycogen in liver and associated **hypoglycemia**.

Fructosuria

- Benign metabolic defect due to deficiency of **fructokinase**.
- There is no abnormality other than excretion of fructose in urine.
- Fructose is not dietary essential.
- Urine gives positive Benedict's and Seliwanoff's tests.
- Incidence is 1 in 130,000 births.

Fructokinase; absent in Fructosuria



- Fructose is phosphorylated by **fructokinase**.
- Fructokinase phosphorylates at **1st position**, whereas hexokinase action is on the 6th position.
- Fructokinase is not dependent on insulin.

GALACTOSE METABOLISM

- Galactose is a constituent of lactose, milk sugar.
- Galactose is not an essential nutrient, because UDP glucose can form UDP galactose.
- Galactose is metabolized almost exclusively by the liver.
- UDP galactose is the active donor of galactose during synthetic reactions.

- Galactose is necessary for synthesis of the following.

i. Lactose Synthesis

Epimerase

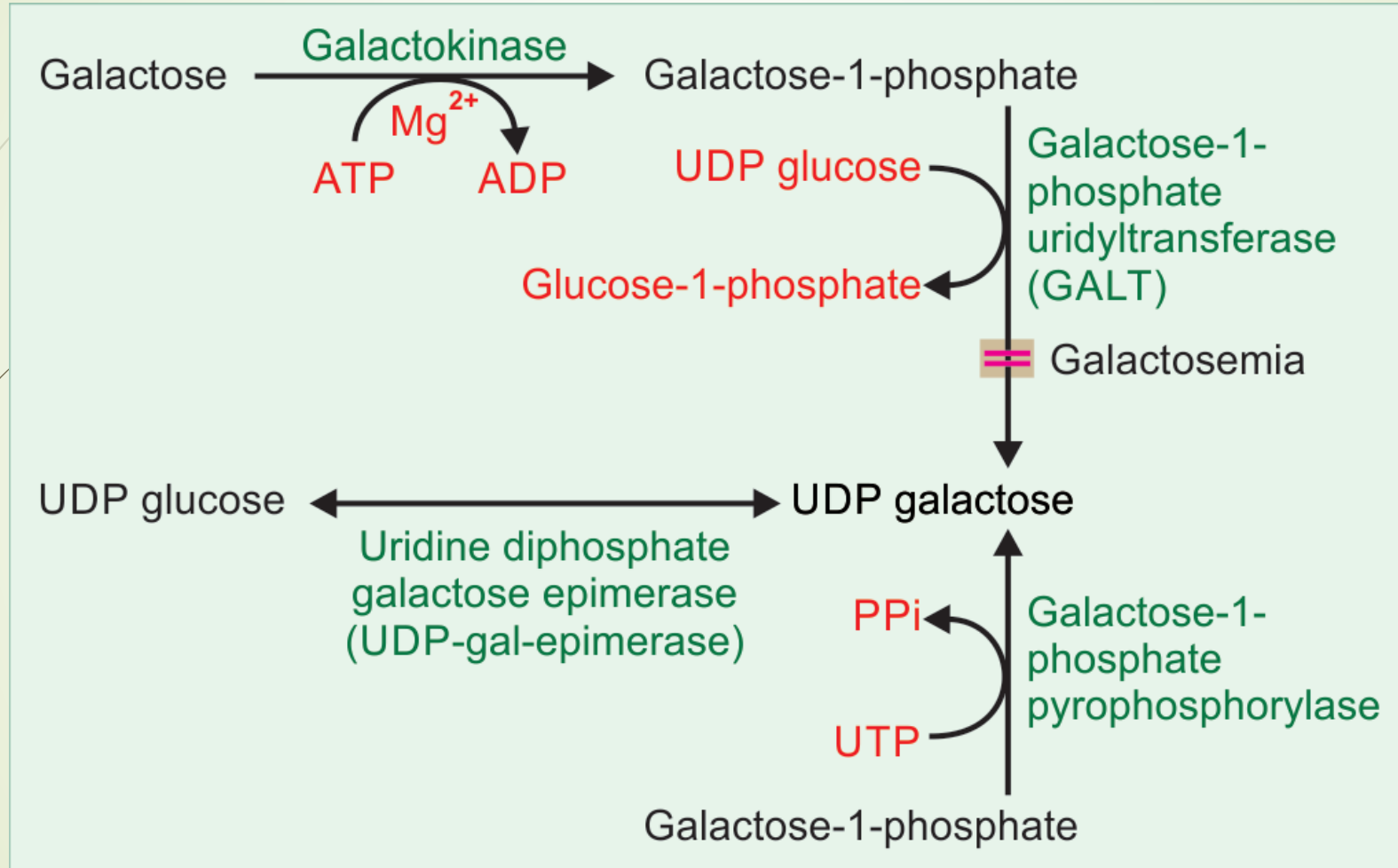
UDP glucose $\longrightarrow$ UDP galactose

Lactose synthase

UDP galactose + glucose $\longrightarrow$ Lactose

Lactose synthesis is seen in lactating mammary glands.

- ii. Synthesis of glycosaminoglycans
- iii. Synthesis of cerebrosides
- iv. Synthesis of glycolipids
- v. Synthesis of glycoproteins.





Alternate pathway: Galactose-1-phosphate pyrophosphorylase in liver becomes active only after 4 or 5 years of life.

- The enzyme will produce UDP-galactose directly which can be epimerized to UDP-glucose.

Galactosemia

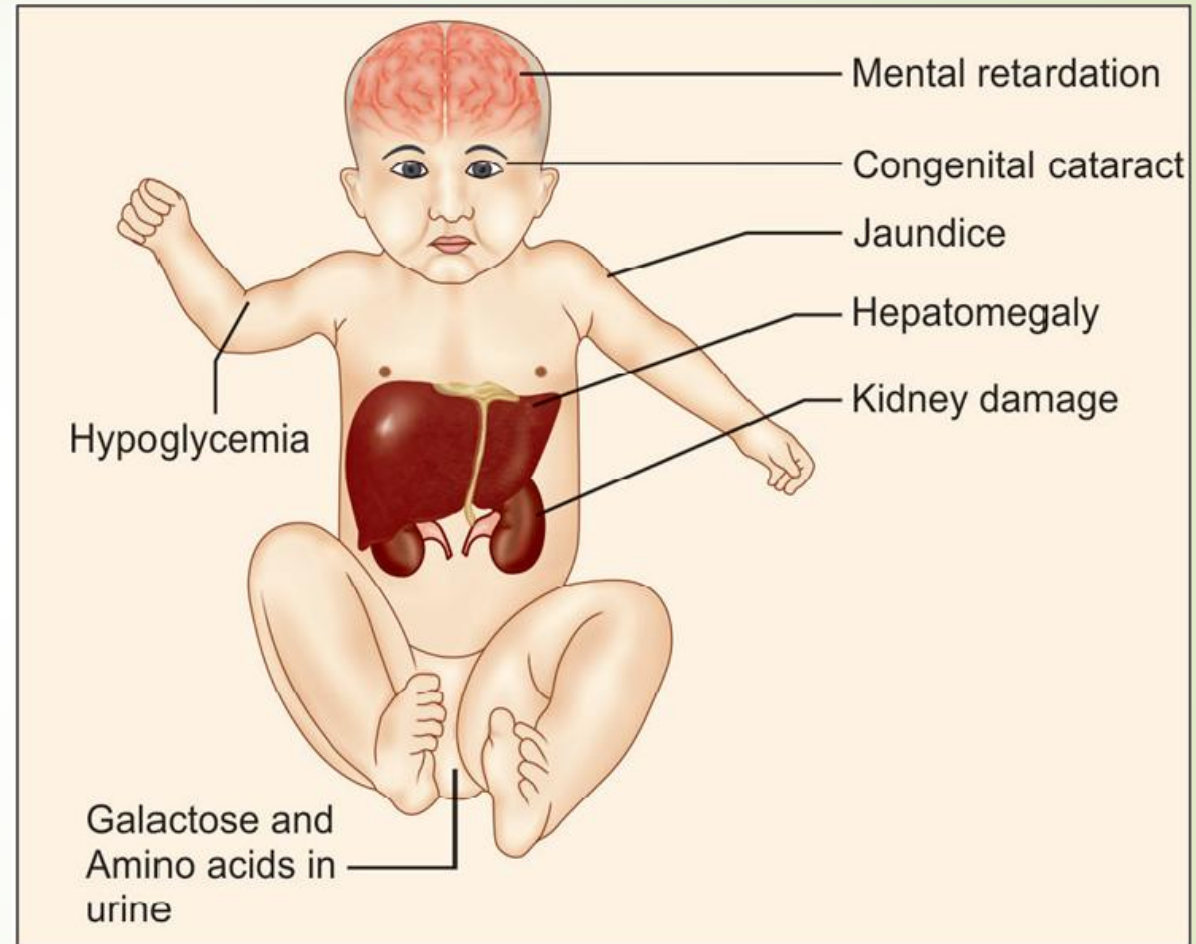
**Inborn error of metabolism.
The incidence is 1 in 35,000 births.**

**Deficiency of the enzyme
galactose-1-phosphate uridyl transferase.**

**Galactose-1-phosphate accumulates
in liver, which inhibits
glycogen phosphorylase.
Neonatal hypoglycemia**

- 
- Bilirubin uptake is less and bilirubin conjugation is reduced; so **unconjugated bilirubin level** is increased in blood.
 - There is enlargement of liver, jaundice and severe **mental retardation**.
 - Free galactose accumulates, leading to **galactosemia**.
 - It is partly excreted in urine (**galactosuria**).
- 

- Galactose is reduced to dulcitol.
- The accumulation of dulcitol in the lens results in cataract due to its osmotic effect.
- This is called **congenital cataract** and is a very characteristic feature of galactosemia.
- Galactose-1-phosphate may get deposited in renal tubules, producing tubular damage leading to generalized **amino aciduria**.



Diagnosis of Galactosemia

Clinical manifestation including congenital cataract and presence of galactose in urine as well as elevated blood galactose levels will help in the diagnosis.

Confirmation by showing deficiency of enzyme activity in WBCs.

Collection of fetal cells by amniocentesis for prenatal diagnosis.

Treatment of Galactosemia

Lactose-free diet.

Special diet may be withdrawn after 4 years, when galactose-1-phosphate pyrophosphorylase becomes active.

**If lactose is withdrawn from the diet,
most of the symptoms recede.**

**But mental retardation, when established,
will not improve.**

Hence early detection is most important.

Galactokinase Deficiency

- A variant of the disease occurs due to the deficiency of galactokinase.
- But here the symptoms are milder.
- This is because galactose-1-phosphate is not formed and hence no toxic effects of this compound are manifested.
- However, cataract is seen.
- Galactokinase deficiency is reported to be 1 in 40,000 births.

Neonatal hypoglycemia

- 
1. Glycogen Storage Disease Type I
 2. Fructose Intolerance
 3. Galactosemia
 4. Medium Chain Fatty Acyl CoA Dehydrogenase deficiency
 5. Long Chain Fatty Acyl CoA Dehydrogenase deficiency
- 